



REDSTONE CSP PROJECT



Report ID	CC IPL2335/VCS/VAL/RCSP/20240624
Project title	Redstone CSP Project
Project ID	5146
Crediting period	30-November-2024 to 29-November-2031
Original date of issue	11-November-2024
Most recent date of issue	19-January-2026
Version	5.2
VCS Standard Version	4.7
Client	Anthesis B.V. ('Anthesis')
Prepared by	Carbon Check (India) Private Limited
Approved by	Amit Anand, CEO

Work carried out by

Campal Deepak Kadam (Team Leader)

Anubhav Dimri (Team Member/ Technical Expert/ Local Expert)

Indumathi C. (Technical Reviewer)

Summary:

- A description of the validation and the project

Anthesis has commissioned Carbon Check (India) Private Ltd. (CC IPL) to validate the project activity “Redstone CSP Project’/A10/. The proposed project activity is a 100 MW concentrated solar thermal power plant. The project utilizes ThermaVault technology which combines solar thermal and molten salt energy storage technologies. Thermal energy is converted to electricity through 115MW high temperature subcritical steam turbine generating units. Prior to implementing this project electricity was supplied by the current fleet of power plants owned and operated by Eskom, the majority of which are coal-based. This CSP plant will reduce greenhouse gas (GHG) emissions by replacing carbon intensive energy with clean renewable electricity. The project is located in Northern Cape province of South Africa. The project is expected to result in an average annual reduction of approximately 478,316 tCO₂e in GHG emissions and a total of 3,348,215 tCO₂e emission reductions over the first crediting period of seven years.

- The purpose and scope of validation

Purpose: The purpose of a validation is to have a thorough and independent assessment of the proposed project activity against the applicable VCS requirements, in particular, the project's baseline, monitoring plan and the project's compliance with relevant VCS and host Party criteria. These are validated in order to confirm that the project design, as documented, is sound and reasonable and meets the identified criteria. Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of emission reductions. Carbon Check's objective is to perform a thorough, independent assessment of the validation of the project activity.

Scope: The validation scope is defined as an independent and objective review of the Project Description (PD). The PD is reviewed against the relevant criteria and guidance documents provided by VCS which included the following: VCS Program Guide, version 4.5, VCS Standard, Version 4.7, Program Definitions, version 4.5, Registration & Issuance Process, version 4.5 and in line with the VCS Validation and Verification Manual, version 3.2/A11/ applicable at the time in order to confirm that the project meets the applicability conditions of the selected baseline and monitoring methodology, ACM0002 version 22.0/A12/ and also assess the claims and assumptions made in the PD without limitation on the information provided by the project proponents.

- The method and criteria used for validation

Validation consists of the following four phases:

- I. A desk review of the project description documents
 - A review of data and information.
 - Cross checks between information provided in PD and information from sources with all necessary means without limitations to the information provided by the project proponent;

II. On-site visit and follow-up interviews with project stakeholders

- Interviews with relevant stakeholders in host country with personnel having knowledge with the project development via telephone, email, or On-site site visits;
- Cross checking between information provided by interviewed personnel with all necessary means without limitations to the information provided by the project proponent;

III. Reference to available information relating to projects or technologies similar to project under validation and review based on the approved methodology being applied for the appropriateness of formulae and accuracy of calculations.

IV. The resolution of outstanding issues and the issuance of the final validation report and opinion.

- The number of findings raised during validation

During the course of validation, a total of 15 were raised, which include:

04 Corrective Action Requests (CARs);

11 Clarification Requests (CLs);

01 Forward Action request (FAR).

All the raised CARs and CLs have been successfully resolved by PP. The FAR raised during the validation shall be addressed in every subsequent verification.

- Any uncertainties associated with the validation

The PD/A1/, emissions reduction calculations/F1/ along with the supporting documents provided are considered to be in line with the VCS version 4 requirements. The validation team has detected no further uncertainties or quality restriction.

- Summary of the validation conclusion

Carbon Check (India) Private Ltd. hereby confirms that the project is fulfilling the criteria specified by VCS PD template version 4.4, VCS Standard Version 4.7, applied methodology ACM0002 version 22.0 and hence be successfully validated under VCS. Carbon Check confirms a positive validation opinion confirming the project complies with the applicable VCS version 4 requirements, thus recommending the project for registration.

CONTENTS

1	INTRODUCTION	6
1.1	Objective	6
1.2	Scope and Criteria	6
1.3	Reasonableness of Assumptions.....	7
1.4	Summary Description of the Project	7
2	VALIDATION PROCESS	7
2.1	Method and Criteria	7
2.2	Document Review.....	8
2.3	Interviews.....	8
2.4	Site Visits	11
2.5	Resolution of Findings.....	12
3	VALIDATION FINDINGS	13
3.1	Project Details	13
3.2	Safeguards and Stakeholder Engagement.....	20
3.3	Application of Methodology	59
3.4	Non-Permanence Risk Analysis.....	84
4	VALIDATION OPINION	84
4.1	Validation Summary	84
4.2	Validation Conclusion.....	85
	APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION.....	87
	APPENDIX 2.1 : REFERENCE DOCUMENTS.....	88
	APPENDIX 2.2 : BACKGROUND DOCUMENTS	90
	APPENDIX 2.3 : CERTIFICATES OF COMPETENCE	91
	APPENDIX 2.4 : FINDINGS LOG	97

1 INTRODUCTION

1.1 Objective

Anthesis B.V. has commissioned the VVB, Carbon Check (India) Private Ltd. to perform validation of the VCS Project Activity “Redstone CSP Project”. The project is a 100 MW concentrated solar thermal power plant located at farm 469 the Hay RD, between the towns of Daniëlskuil and Postmasburg, in the Northern Cape Province of South Africa. This report summarises the findings of the validation of the project, performed on the basis of the VCS Program Guide, version 4.5, VCS Standard, version 4.7, VCS Program Definitions, version 4.5, Registration & Issuance Process, version 4.5 and VCS Validation and Verification Manual, version 3.2.

The purpose of a validation is to have a thorough and independent assessment of the proposed project activity against the applicable VCS requirements, in particular, the project’s baseline, monitoring plan and the project’s compliance with relevant VCS and host Party criteria. These are validated in order to confirm that the project design, as documented, is sound and reasonable and meets the identified criteria. Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of emission reductions, VCU.

1.2 Scope and Criteria

The validation scope is defined as an independent and objective review of the Project Description (PD), project design, the project’s baseline study and monitoring plan and other relevant documents. The PD is reviewed against the relevant criteria and decisions by the VCSA, including the approved baseline and monitoring methodology. Carbon Check has employed a risk-based approach in the validation, focusing on the identification of significant risks and reliability of project monitoring and generation of emission reductions.

The validation is not meant to provide any consulting towards the project proponents. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

The validation is carried out on the basis of the following requirements, applicable for this project activity:

- VCS Program Guide (v4.5)
- VCS Standard (v4.7)
- Program Definitions (v4.5)

- Registration & Issuance Process (v4.5)
- VCS Validation and Verification Manual (v 3.2)
- CDM Methodology: ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0)
- Other relevant rules, including the host country legislation.

1.3 Reasonableness of Assumptions

☒ Reasonable level of assurance

☐ Limited level of assurance

1.4 Summary Description of the Project

The proposed project activity 'Redstone CSP Project' is a 100 MW concentrated solar thermal power plant. The project utilizes Therma Vault technology which combines solar thermal and molten salt energy storage technologies. Thermal energy is converted to electricity through 115MW high temperature subcritical steam turbine generating units. Prior to implementing this project electricity was supplied by the current fleet of power plants owned and operated by Eskom, the majority of which are coal-based. This CSP plant will reduce greenhouse gas (GHG) emissions by replacing carbon intensive energy with clean renewable electricity. The project is located in Northern Cape province of South Africa. The project is expected to result in an average annual reduction of approximately 478,316 tCO₂e in GHG emissions and a total of 3,478,316 tCO₂e emission reductions over the first crediting period of seven years.

2 VALIDATION PROCESS

2.1 Method and Criteria

Anthesis B.V. has commissioned the VVB, Carbon Check (India) Private Ltd. to perform validation of the VCS Project Activity "Redstone CSP Project", with regards to the relevant requirements of VCS Standard, Version 4.7/A11/.

The validation includes a thorough and independent assessment of the proposed project activity against the applicable VCS requirements, in particular, the project's baseline, monitoring plan and the project's compliance with relevant VCS and host Party criteria. The validation involves assessment of the project and to confirm that the project meets the applicability conditions of the selected baseline and monitoring methodology, ACM0002, Version 22.0 and also assess the

claims and assumptions made in the PD without limitation on the information provided by the project proponents. The overall validation was conducted using CCIPL's internal procedures.

2.2 Document Review

The VCS project description, emission reduction calculation spread sheet and supporting documents related to the project design and baseline were reviewed as per VCS Standard, Version 4.7 standard requirements. The desk review included:

- A review of the data and information presented to verify completeness and consistency in accordance with VCS version 04 requirements;
- A review of the project description and monitoring methodology, paying particular attention to the applicability conditions of the methodology and baseline and additionality related requirements.
- A review of the monitoring plan and the project's compliance with relevant VCS criteria.

Furthermore, the validation team used additional documentation by third parties like host-party legislation, technical reports referring to the project design or to the basic conditions and technical data.

2.3 Interviews

An on-site visit to the project activity was undertaken on 07-November-2024 to confirm the information as outlined in the table below and to resolve issues identified in the document review. The on-site visit was conducted to assess the implementation and operation of the project activity and to review evidence, and interview key personnel to confirm evidence associated with the project design, implementation, plant operations, environmental impacts, stakeholders etc.

Sr.	Date	Name	Organisation	Topic	Interviewed by
/01/	07-November-2024	Brigitte Nagel	Anthesis	• Project Design • Project start date and Project Location • Baseline Scenario • Baseline Identification and Additionality • Monitoring and reporting documentation	Campal Kadam and Anubhav Dimri

				• Quality Assurance – Management and operating system • Social and Environmental Impacts • Compliance with relevant laws	
/02/	07-November-2024	Grant Little	Anthesis	• Project Design • Project start date and Project Location • Baseline Scenario • Baseline Identification and Additionality • Monitoring and reporting documentation • Quality Assurance – Management and operating system • Social and Environmental Impacts • Compliance with relevant laws	
/03/	07-November-2024	Tshidaho Mapande	ACWA	• Project Design • Project start date and Project Location • Baseline Scenario • Baseline Identification and Additionality • Monitoring and reporting documentation • Quality Assurance – Management and operating system • Social and Environmental Impacts • Compliance with relevant laws	

/04/	07-November-2024	Miguel Poblete	ACWA – Redstone CSP plant manager	• Project Design • Project Implementation status • Project start date and Project Location • Qualification and Training • Plant Operations • Monitoring and reporting documentation • Quality Assurance – Management and operating system • Social and Environmental Impacts • Compliance with relevant laws	
/05/	07-November-2024	Pulane Mogoere	ACWA	• Project Design • Project start date and Project Location • Baseline Scenario • Baseline Identification and Additionality • Monitoring and reporting documentation • Quality Assurance – Management and operating system • Social and Environmental Impacts • Compliance with relevant laws	
/06/	07-November-2024	Helena English	Local Stakeholder and Mayor of Tsantsabane local municipality.	• Local Stakeholder Consultation • Social and Environmental Impacts	

/07/	07-November-2024	Kealeboga T	Local Stakeholder and Councilor of Tsantsabane local municipality.	- Local Stakeholder Consultation - Social and Environmental Impacts	
/08/	07-November-2024	L.C. Sehako	Local Stakeholder and personnel at Department of Education	- Local Stakeholder Consultation - Social and Environmental Impacts	
/09/	07-November-2024	Norton S	Local Stakeholder and personnel at Department of Education	- Local Stakeholder Consultation - Social and Environmental Impacts	

2.4 Site Visits

As discussed in the above section 2.3, an on-site visit was undertaken by the validation team on 07-November-2024 to carry out the following; -

- An assessment of the project design and technical specification, project location, project boundary, additionality, baseline scenario, baseline methodology, GHG emissions quantification and implementation status and operation of the project activity as per the PD.
- A review of information flows for generating, aggregating and reporting the monitoring parameters and monitoring methodology.
- Interviews with relevant personnel to determine whether the operational and data collection procedures are implemented in accordance with the monitoring plan in the PD;
- The assessment team has verified sufficient appropriate audit evidences, to reduce audit risk to an acceptably low level as requisite to achieve reasonable level of assurance for the current validation.

- Interview local stakeholders to validate the local stakeholders consultation process and understand the potential benefits and negative impacts of the project activity.

2.5 Resolution of Findings

This section summarizes the findings from the validation of the project activity. In this section the findings from the document review, site visit, assessments and interviews are provided.

Material discrepancies identified in the course of the validation are addressed either as CARs, CLs or FARs.

Corrective action requests (CAR) are issued, where:

- i. mistakes have been made with a direct influence on project results requiring adjustments of the VERs/VCUs monitoring report;
- ii. applicable methodological specific requirements have not been met.

A Clarification request (CL) may be used where additional information is needed to fully clarify an issue or where the information is not transparent enough to establish whether a requirement is met.

A total of 04 CARs and 11 CLs have been raised for the validation of the project activity. All the raised findings have been successfully resolved by PP. Please refer to Appendix 4 below for the details of the CARs/CLs and their closure.

2.5.1 Forward Action Requests

A forward action request (FAR) should be issued, where:

- i. the actual project monitoring and reporting practices requires attention and /or adjustment for the next consecutive verification period, or
- ii. an adjustment of the MP is recommended.

In the context of FARs, risks have been identified, which may endanger the delivery of high-quality emissions reductions in the future, i.e. by deviations from standard procedures as defined by the MP. As a consequence, such aspects should receive a special focus during the next consecutive verification. A FAR may originate from lack of data sustaining claimed emission reductions.

01 FAR has been raised during the course of validation which shall be addressed in every subsequent verification.

3 VALIDATION FINDINGS

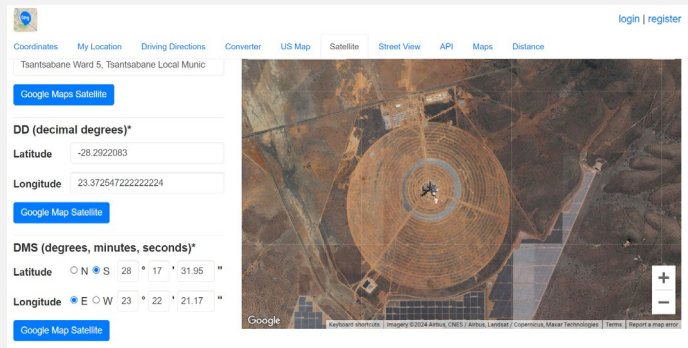
3.1 Project Details

The validation involves assessment of the project and to confirm that the project meets the applicability conditions of the selected baseline and monitoring methodology ACM0002 version 22.0 and the project description is accurate and complete as per the VCS version 4 requirements.

Item	Evidence gathering activities, evidence checked, and assessment conclusion				
Audit history	Audit type	Period	Program	Validation/verification body name	Number of years
	Val	October 2024	VCS	Carbon Check (India) Private Limited	--
Sectoral scope	01				
AFOLU project category, if applicable	Not applicable as the proposed project activity is a non-AFOLU project				
Project activity type	Energy (Renewable/Non- Renewable)				
General eligibility of the project to participate in the VCS Program	- The project is not excluded from the VCS program as demonstrated in section 1.4.1 of the PD. As per section 2.1.3 of the VCS Standard v4.7, the exclusion for " Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants. " does not apply to concentrated solar thermal-to-electricity, floating solar PV, or energy storage systems (e.g., batteries). The VVB based on the review of the project design, project technology and visit to the project location confirms that the proposed project activity is a concentrated solar thermal to electricity technology and thus eligible under the VCS program. - The project activity was listed on the VERRA registry on 18-September-2024. The public commenting period for the activity lasted from 18-September-2024 to 18-October-2024. VVB has confirmed the same through review of the project page on VERRA registry. https://registry.verra.org/app/projectDetail/VCS/5146) VVB submitted the NOVS for validation of the project activity on 01-October-2024.The opening				

	meeting with the VVB was carried out on 07-November-2024 which is 15 business days after submission of NOVS. The validation is expected to complete during December 2024 which is within 2 years of the expected start date of crediting period thus meeting the validation deadline. - Based on review of the project description/A1/, VVB confirms that the applied methodology is eligible under the VCS Program. The applied methodology, ACM0002 version 22 is a large scale consolidated CDM methodology and can be applied to large scale projects in VCS. - The project is not a fragmented part of a larger project or activity that would otherwise exceed such limits.
AFOLU project eligibility, if applicable	Not applicable as the proposed project activity is a non-AFOLU project.
Transfer project eligibility, if applicable	Not applicable as the proposed project activity is not a transfer project and did not participate under any other GHG program. Declaration stating the same is provided to the VVB by PP.
Project design	The proposed project activity is a standalone project and does not involve multiple instances.
Project ownership	ACWA Power Solar Reserve Redstone Solar Thermal Power Plant (RF) (Pty) Ltd has the legal right to own and operate the Redstone CSP facility. VVB has verified the following documents/B5/ to confirm the legal ownership of the project activity: - IPP license - Lease direct agreement dated 06 December 2019 - Formalities certificate – Landowner Further, VVB has reviewed the joint ownership agreement of carbon credits to confirm that there exists a legal agreement between ACWA Power Solar Reserve Redstone Solar Thermal Power Plant (RF) (Pty) Ltd and Anthesis B.V. regarding joint ownership of carbon credits generated through the project activity/D1/.

	Both the entities have been clearly and sufficiently described in section 1.6 of the PD and provisions regarding ownership of project activity are described in section 1.8 of the PD.
Project start date	The expected start date of the project activity is 30-November-2024 which is the date the project activity is expected to be commissioned.
Project crediting period	The expected start date of crediting period of the project activity is 30-November-2024 which is the date the project activity is expected to be commissioned and start generating emission reductions/A2/.
Project scale	The project is expected to generate annual average emission reductions of 478,316 tCO ₂ e. Therefore, the project is classified as large project in section 1.11 of the PD which in line with § 3.10.1 of the VCS standard.
Likelihood of achieving estimated GHG emission reduction or removals	VVB has reviewed the project design and cross-checked the estimated emission reductions by reproducing the ER calculations. VVB has reviewed the assumptions, methods and limitations involved in estimating the emission reductions and found them to be reasonable. Therefore, VVB confirms that the project activity is likely to achieve the estimated GHG emissions reductions.
Technologies and measures implemented by the project activity or activities	The project utilizes Therma Vault technology which combines solar thermal and molten salt energy storage technologies/C1/. Solar energy is captured via a circular heliostat field made up of 41,260 heliostats (each with 25 m² mirror reflective surface area). The heliostats track the sun and reflect the sunlight onto a 250 m tall central receiver tower, which uses molten salt as heat transfer and storage medium. Thermal energy is converted to electricity through 115MW high temperature subcritical steam turbine generating units. The plant lifetime is expected to be 25 years. VVB has reviewed technical specifications of following auxiliary components of the CSP plant. • Solar collected field : To confirm that surface area of individual collector field is 25m². • Receiver : A 250-meter-tall concrete tower and thermal receiver rated at 600 MW thermal (MWth).

	• Cold and hot storage tanks: The cold salt storage tank holds the cold molten salts (300°C), from where it is pumped to the receiver where the salt is heated to 565 °C. • Power block: A thermal to electric power block, with a 115 MWe gross power and 100Mwe net power at design conditions. • Water treatment plant which includes Ultra filtration, Reverse osmosis, Electro deionization (EDI) System, Chemical dosing
Implementation schedule of the project activity or activities	At the time of physical site visit to the plant, VVB is able to verify that the project is currently in the commissioning stage waiting to obtain a certificate of commission. Thus, the project is expected to start operation as per the implementation schedule and no significant delays are anticipated.
Project location	The project is located in Northern Cape province of South Africa. The physical address as of the plant as verified by VVB during the physical site visit to the CSP plant is as follows : Farm 469, the Hay District (Administration District), Tsantsabane Local Municipality of the ZF Mcgawu District. The coordinates verified are as follows : 
Conditions prior to project initiation	Assessment of conditions prior to project initiation is included in section 3.4 9 'Baseline scenario' of this report.
Project compliance with applicable laws, statutes and other regulatory frameworks	PP has identified and described relevant legislations that the project will comply with in section 1.15 of the PD. Based on its own review of legislations applicable in host country South Africa and interview with representatives of PP, VVB confirms

	that the legislations identified are sufficient and PP will adhere to these legislations during the crediting period. The identified and verified legislations are as follows : • Constitution of the Republic of South Africa (Act 108 of 1996) • National Development Plan, Agenda 2030 • Electricity Regulation Act (Act 4 of 2006) and the Electricity Regulation on New Generation Capacity R399 of 4 May 2011, as amended by R1366 of 04 November 2016 • Integrated Resource Plan (IRP2019), and the draft IRP2023 • National Environmental Management Act, (NEMA) (Act No. 107 of 1998) • National Environmental Management Biodiversity Act, (NEMBA). (Act No. 107 of 1998) • Northern Cape Nature Conservation Act. (Act No. 9 of 2009) • Conservation of Agricultural Resources (Act 43 of 1983) • National Water Act (No. 36 of 1998) • Occupational Health and Safety Act (No. 85 of 1993) • The Basic Conditions of Employment Act (No. 75 of 1997) • Carbon Tax Act 15 of 2019 and supporting regulations
Double counting and participation under other GHG programs	The VVB has checked for double counting at the validation stage by reviewing all the relevant registries that could hold RECs from the project. The project activity is not registered under any other emissions trading program or any other mechanism that includes GHG allowance trading. Following registries have been assessed: • https://www.climateactionreserve.org/ • https://registry.goldstandard.org/projects?q=&page=1 • https://iceland.itmoregistry.net/Public/Project • https://biocarbonregistry.com/en/projects/

	• https://projects.globalcarboncouncil.com/pages/submitted_projects • https://registry.verra.org/app/search/VCS/All%20Projects • https://cdm.unfccc.int/Projects/projsearch.html • The project is not registered under the international REC Mechanism e.g. I-REC Device Registry and the same is confirmed through the I-REC web site (https://v-1.evident.app/Public/ReportDevices/). The project has provided all information required on double counting and double claiming in section 1.16 and 1.17 of the VCS PD. VVB has confirmed through interviews with representatives of PP that the project is not claiming and does not plan to claim GHG credits under any other GHG program. Further, PP has confirmed that the project activity has not been rejected under any other GHG program. A signed declaration stating the same is provided by PP to the VVB.
No double claiming with emissions trading programs or binding emission limits	The VVB has checked for double counting at the validation stage by reviewing all the relevant registries that could hold RECs from the project. The project activity is not registered under any other emissions trading program or any other mechanism that includes GHG allowance trading. Following registries have been assessed: • https://www.climateactionreserve.org/ • https://registry.goldstandard.org/projects?q=&page=1 • https://iceland.itmoregistry.net/Public/Project • https://biocarbonregistry.com/en/projects/ • https://projects.globalcarboncouncil.com/pages/submitted_projects • https://registry.verra.org/app/search/VCS/All%20Projects • https://cdm.unfccc.int/Projects/projsearch.html • The project is not registered under the international REC Mechanism e.g. I-REC Device Registry and the same is confirmed through the I-REC web site (https://v-1.evident.app/Public/ReportDevices/).

	The project has provided all information required on double counting and double claiming in section 1.16 and 1.17 of the VCS PD. VVB has confirmed through interviews with representatives of PP that the project is not claiming and does not plan to claim GHG credits under any other GHG program. Further, PP has confirmed that the project activity has not been rejected under any other GHG program. A signed declaration stating the same is provided by PP to the VVB/A4/.
No double claiming with other forms of environmental credit	The response should include: Where the project activity has sought, received, or is planning to receive credit from another GHG-related environmental credit system, assess the evidence of no double claiming against VCS Standard requirements.
Supply chain (Scope 3) emissions double claiming	Based on review of the proposed project design, VVB confirms that the project proponent(s) or authorized representative is not a buyer or seller of a product whose emissions footprint is changed by the project activities. Same is clearly indicated in section 1.17.3 of the PD.
Sustainable development contributions	PP has identified three SDG contributions achievable through implementation of the project , assessment of which is as follows : - The project activity will contribute towards SDG target 7.2 through increase in amount of renewable energy in South Africa. The activity aims to displace 10,500 GWh of emission intensive grid electricity through generation of low emission CSP electricity. Based on review of the proposed project design, technology involved, and the monitoring plan, VVB confirms that the SDG target is relevant and applicable to the project activity and the SDG indicator chosen (7.2.1) is appropriate. VVB concludes that the estimated contribution over lifetime is likely to be achievable and can be monitored. - The project activity will contribute towards SDG target 13.0 through avoidance of approximately 10,478,316 tCO2e over the project lifetime. Based on review of the proposed project design, technology involved, and the monitoring plan, VVB confirms that the SDG target is

	relevant and applicable to the project activity and the SDG indicator chosen is appropriate. VVB concludes that the estimated contribution over lifetime is likely to be achievable and can be monitored. The project activity will contribute towards SDG target 8.5 through creating of 481 jobs during the construction phase and 41 jobs during the operational phase. Based on review of the proposed project design and interview with management of the CSP plant, VVB confirms that the SDG target is relevant and applicable to the project activity and the SDG indicator chosen is appropriate. VVB concludes that the estimated contribution over lifetime is likely to be achievable and can be monitored.
Additional information relevant to the project	The response should include: - As per the applied methodology ACM0002 version 22, leakage emissions are considered to be '0'. Therefore, the risk of leakage is negligible to nil. - No commercially sensitive information that has been excluded from the public versions of project documents. - VVB confirms that based on review of the project design and overall assessment of the project activity, VVB did not find any additional relevant information that may have a bearing on the eligibility of the project, the reductions or removals, or the quantification of the project's reductions or removals.

3.2 Safeguards and Stakeholder Engagement

3.2.1 Stakeholder Engagement and Consultation

3.2.1.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder identification	The VVB, through review of PD/A1/ and interviews with representatives of PP confirms that sufficient mechanism is in place

	to identify stakeholder, and PP has identified relevant stakeholders as described in the PD.
Legal or customary tenure/access rights	The VVB confirms that project activity is greenfield CSP plant constructed on leased land. Therefore, there are no legal or customary tenure/access rights to territories and resources, including collective and conflicting rights, held by stakeholders, indigenous people (IPs), local communities (LCs), or customary rights holders which can be conflicted due to the distribution of the clean energy products. Hence, not applicable.
Stakeholder diversity and changes over time	VVB confirms that implementation of project is as described in the VCS PD. The stakeholders identified have not changed over time and reassessment of stakeholder diversity is not required during the current monitoring period. The relevant stakeholders have been sufficiently described in the PD.
Expected changes in well-being	The project activity leads to the economic and social well-being of stakeholders by creating local employment opportunities in operational, management, manufacturing, distribution, and logistics roles. This has been confirmed by VVB through PP and stakeholders interviews during on site audit.
Location of stakeholders	The VVB confirms that stakeholders impacted by the grouped project include the people residing in the municipality and those who are getting employment due to the project activity. Hence, the impact of the project devices is deemed localized and hence areas outside the project area are not predicted to be impacted by the project.
Location of resources	The VVB confirms that the project activity is CSP plant constructed on private leased land and hence does not interfere with rights to lands, territories and/ or resources.

3.2.1.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder engagement process	Stakeholder consultation was conducted between August – December 2011 and August-2024.

Consultation outcome	No adverse comments were received for both the meetings, and this is addressed in the PD. This was also confirmed by the validation team during the on-site interviews.
Ongoing communication	The mechanism for ongoing communication with local stakeholders has been described in the PD. VVB has visited the grievance box located in the local municipality office to confirm the same.
Stakeholder input	VVB confirms that no changes to the project design have been made owing to the positive reception of the project as confirmed during the on-site audit.

3.2.1.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Obtaining consent	VVB has confirmed through review of land lease agreement that the owner has consented to the project activity being constructed and operational on the property. Additionally, VVB has interviewed the local stakeholders particularly the Mayor of the municipality under which the project is implemented to confirm that the project activity does not impact customary rights of Indigenous people and local communities.
Outcome of FPIC discussion	Based on review of the evidence and interviews with representatives of PP and relevant local stakeholders, VVB confirms that VVB did not find any evidence such that the proposed project activity would negatively impact free, prior and informed consent of stakeholders involved.

3.2.1.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Development process	VVB has reviewed the grievance redressal procedure and confirms that the mechanism is sufficiently described in 2.1.4 of the PD. VVB has validated the mechanism designed by checking the following : 1. The grievance portal on the Anthesis website 2. Grievance boxes placed at local municipality office

Grievance redress procedure	VVB confirms that the procedure established by PP to receive and address the feedback is sufficiently described in section 2.1.4 of the PD. The mechanism is sufficiently designed as per VCS requirements.
-----------------------------	---

3.2.1.5 Public Comments

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
The project activity was open for public comments on VERRA registry from 08-September-2024 to 08-October-2024. No public comments were received during the period.	--	VVB confirms that no public comment was received during the public commenting period as communicated by VERRA Secretariat to PP through an email dated 18-October-2024.

PD has received comments after 7 months of completion of initial public commenting period. The same is explained in section 2.15 of the PDD:

Comments received	Actions taken	VVB Assessment
Comment 1 – Anonymous – submitted 16 May 2025: It has recently come to our attention that several projects listed below have recently completed their public stakeholder consultation, along the process of applying for registration to generate carbon credits under the VCS. 1. Scatec RMI Hybrid Cluster (https://registry.verra.org/app/projectDetail/VCS/5488)	Stakeholder engagement for the project followed the requirements stipulated in §3.18 of the VCS Standard. This included building on the extensive stakeholder process undertaken during the EIA phase of the project and augmenting this through a renewed review of the stakeholder database to identify and add to it, confirming any stakeholders with legal or customary rights tenure or access rights or any conflicting rights. The stakeholder engagement process included physical	VVB has reviewed the public comment and the clarification provided by the PP and confirms that it is appropriate based on the stakeholder consultation records/documents listed in Appendix 2 of the validation report. To further verify this, the VVB interviewed stakeholders (listed in Section 2.3 of this report) and verified the notification posted for continuous stakeholder consultation at site during the on-site audit. Therefore, the VVB finds that the Local Stakeholder

2. De Aar Hydra Storage (https://registry.terra.org/app/projectDetail/VCS/5337) 3. Oya Hybrid Project (https://registry.terra.org/app/projectDetail/VCS/5312) 4. Umoyilanga Hybrid Project (https://registry.terra.org/app/projectDetail/VCS/5272) 5. Redstone CSP project (https://registry.terra.org/app/projectDetail/VCS/5146)	notices located in public areas (municipal buildings, local supermarkets etc.) near the project activity, meetings with local stakeholders, advertisements in newspapers, electronic (email) communication, some telephonic calls and a dedicated website link is maintained (https://www.anthesisgroup.com/za/project-comments/) that remains open for stakeholder concerns, complaints, grievances and engagement. As part of the operational phase, each project also has a dedicated team of specialists located on site specifically focused on stakeholder engagement. A 30-day Global Stakeholder Consultation (GSC) managed on the Verra registry website was also part of the process as per §3.18.9 – 3.18.13 of the stakeholder engagement process. The stakeholder engagement process was validated and checked by the independent VVB during the validation process, and included verification of documentation and communication, interviews with the project team members who were involved in the implementation and interviews with a sample of stakeholders themselves.	Consultation (LSC) meets the requirements as per §3.18 of the VCS Standard, Version 4.7.
We have not received any notification from any of the project developers of their intention to pursue registration. As a significantly interested and affected stakeholder, we therefore object to the registration and request further consultation on several issues of concern.		

	The comments received from VERRA were anonymised, so it was unsure who the stakeholder was in order to review any possible comments or concerns raised during the engagement process. As part of our ongoing stakeholder engagement (as per §3.18.5 – 3.18.6 of VCS Standard) stakeholder may engage, comment, raise a grievance or be added to the stakeholder database. Various mechanisms are available, the primary ones being through the project operational stakeholder teams or via the weblink (https://www.anthesisgroup.com/za/project-comments/). While the stakeholder engagement process sought to identify and include all relevant stakeholders, any stakeholders missed, still have the opportunity to engage and be part of the ongoing process by accessing the weblink above.	
Comment 2 – Anonymous – submitted 17 May 2025 Projects 5488, 5337, and 5146 have all been awarded tenders under South Africa’s REIPPPP and were procured by Eskom through government-backed bidding rounds. The tariff structure is designed to ensure the bankability and long-term	The response to the comment has been split into the primary issues raised as follows: • Additionality • Ownership of project/Vcus • Potential for double or triple counting • Reflection in national Grid Emission Factor	VVB has reviewed the public comment and assessed the following aspects of project concerning the comment: 1. Additionality : The comment raises a concern that additionality of the project activity may be impacted due to the tariff structure. VVB has assessed the additionality of

viability of the projects. These are large-scale renewable energy projects with significant credit generation potential. We reached out to Eskom and confirmed that the renewable energy from these projects is counted toward the national grid emission intensity. This raises several concerns, including; additionality, ownership, double or triple counting - reflected in the national grid factor, at least one project is selling electricity to a steel manufacturer that is using the green power to reduce its emissions intensity, raising the possibility of multiple claims on the same environmental benefits. There's also a stakeholder engagement concern - one project listed Eskom as having been engaged in April, but Eskom later indicated they were unaware of any such engagement. There could also be potential (non-public) legal disputes related to the projects.	• Potential selling green power to a non-Eskom offtaker • Engagement with Eskom as a stakeholder • Potential for (non-public) legal disputes ADDITIONALITY: In accordance with the VCS Standard (§3.14) and the applied methodology (§5.2.3 and the associated Tools that were applied), additionality was evaluated within the framework applicable to the Verra VCS requirements, including CDM TOOL1 v7, CDM TOOL 24 v3.1 and CDM TOOL 27 v14. All parameters used in the assessment of additionality were valid at the time of investment decision, and were verified by the carbon project developer as well as the independent VVB before being submitted to Verra as part of the Registration process. Even though the REIPPPP is a program implemented by the South African government, it is not a government subsidised program, but rather a price-competitive procurement programme. We acknowledge that REIPPPP tariffs are structured to provide cost recovery. However, the assumption that all awarded	this project activity (5146) and confirms that the project activity is automatically additional as demonstrated in section 3.5.2 of the VCS PD. 2. Ownership of project VCU's : VVB has reviewed the license issued by IPP/B2/ to the project proponent ACWA Power Solar Reserve Redstone Solar Thermal Power Plant (RF) (Pty) Ltd. Therefore, the legal ownership of the project activity is confirmed through review of the same. Further, VVB has reviewed an agreement between the PPs ACWA Power Solar Reserve Redstone Solar Thermal Power Plant (RF) (Pty) Ltd (Redstone') in whose name the IPP is issued and Anthesis B.V. ('Anthesis') who are involved in developing the project under VERRA. Based on this agreement, VVB confirms that both parties will have joint ownership of the
--	---	---

	projects are fully bankable without carbon revenue is not borne out in practice. Of the projects commented on, only the Redstone CSP project is a REIPPP project. The other projects were all granted licenses under the Risk Mitigation IPP Programme. Despite this subtle difference, the projects still had to demonstrate additionality in compliance with the Verra rules, irrespective if they were part of the REIPPP programme or not. Additionally, risks such as capex overruns, long development lead times, logistics issues at remote greenfield sites and currency (ZAR) devaluation and volatility impacted financial viability, reinforcing the importance of a carbon revenue stream. The technology applied for the Redstone CSP is on a positive list, so additionality is automatic in terms of the current Verra guidelines. As such, the projects meet the criteria for additionality as per VCS rules. Ownership of project/Vcus: In accordance with the VCS Standard (§3.7), project proponents need to demonstrate the legal right to control and operate the project activities. This and the proof of	carbon project when registered. 3. The VVB has checked for double counting at the validation stage by reviewing all the relevant registries that could hold RECs from the project. The project activity is not registered under any other emissions trading program or any other mechanism that includes GHG allowance trading. Following registries have been assessed: ➤ https://www.climateactionreserve.org/ ➤ https://registry.goldsstandard.org/projects?q=&page=1 ➤ https://iceland.itmoregistry.net/Public/Project ➤ https://biocarbonregistry.com/en/projects/ ➤ https://projects.globalcarboncouncil.com/pages/submitted_projects ➤ https://registry.terra.org/app/search/VCS/All%20Projects ➤ https://cdm.unfccc.int/Projects/projectsearch.html
--	---	--

	right were demonstrated to the VVB during the validation through legally binding agreements that the various project proponents had signed. These agreements covered commercial aspects of the project, including responsibilities for various project activities (such as operations, offtake, carbon development etc.). While the comment suggests a contractual obligation limiting the ownership of the carbon credits to the project proponent, independent legal opinion and review obtained by both project proponents has a differing opinion in this, as the relevant clauses of both the PPA and the Implementation Agreements (IA) are ambiguous regarding the ownership. Where a contract is ambiguous, then one can rely on general principles of law to interpret the contract, and in this instance reliance can be placed on the general principle of property law that the (carbon) project owner (the IPP) is also owner of the environmental attributes with the right to monetise such environmental attributes. Legal review was conducted in the light of these comments and a letter to the IPP from the IPPO dated 1 July 2025. The review was unable to identify any substantive contractual	➤ The project is not registered under the international REC Mechanism e.g. I-REC Device Registry and the same is confirmed through the I-REC web site (https://v-1.evident.app/Public/ReportDevices/). The project has provided all information required on double counting and double claiming in section 1.16 and 1.17 of the VCS PD. VVB has confirmed through interviews with representatives of PP that the project is not claiming and does not plan to claim GHG credits under any other GHG program. Further, PP has confirmed that the project activity has not been rejected under any other GHG program. A signed declaration stating the same is provided by PP to the VVB. 4. Reflection of project activity in national grid emission factor: VVB has reviewed the grid emission factor calculation demonstrated in Grid
--	---	---

	foundation that differs from the PP's current understanding and interpretation of the contracts and the eligibility to register a project and gain VCUs from the project. Furthermore, these contractual obligations also made provision for changes to the operating environment in the future that may arise from changes in ownership, legal disputes, legislation, contractual amendments, policy changes or changes to the international voluntary carbon market, which the PP acknowledges may change any future claim or eligibility. These documents formed part of the independent validation by the VVB during the audit process. Potential for double or triple counting: In accordance with the VCS Standard (§3.23), project proponents need to demonstrate that there is no double counting, or double selling of any GHG emission reductions. This was confirmed by cross referencing a number of local and international registries, including Renewable Energy Certificates (RECs). Further to this, during the project development process, each project owner provided a signed declaration that no other environmental benefits are	Emission factor calculation sheet/F2/. Based on the review, VVB confirms that the EF is calculated as per CDM TOOL 07. All the data sources referenced to determine the EF are found to be relevant, credible and latest. Therefore, VVB confirms that PP has determined the grid emission factor appropriately and sufficiently justified the sources. 5. Selling of renewable energy generated by the project activity: Based on the review of the PPA between the PP and ESKOM/B1/, VVB confirms that PP will sell the renewable energy generated through Redstone project activity only to ESKOM as per terms of agreement outlined in the PPA. Further, VVB based on on-site inspection confirms that there is no other grid present in proximity of project area at the time of validation site visit.
--	---	---

	sought, and no offtake agreements or GHG benefits have been allocated outside of the listed Verra project. These documents formed part of the independent validation by the VVB during the audit process. Reflection in national Grid Emission Factor: Under the VCM, South Africa does not have a published and approved national grid emission factor or standardised baseline, and each project has to determine the Grid Emission Factor in accordance with the VCS Standard (§3.13) and the applied methodology. In the case of this project, CDM methodology ACM 0002 (v 22.0) was applied, and as per §5.2. CDM TOOL07 were applied in order to determine the grid emission factor. This process is approved by Verra and takes cognisance of all components of the national grid, including renewable energy projects. As per the prescribed tools, the projects seeking registration with Verra, as well as other REIPPP projects not yet commissioned at the time of publishing the PDD are not included in the baseline scenario and hence are excluded from the grid emission factor calculation. However, the	6. Engagement of ESKOM as stakeholder: VVB has reviewed the local stakeholder consultation records of the project activity and confirms that the stakeholders have been identified as per §3.18.1 and LSC meetings were carried out prior to the validation as per §3.18.2 of VCS standard v 4.7. Additionally, VVB has cross-checked PP's correspondence with ESKOM to confirm involvement of ESKOM as a stakeholder during construction phase/A.7/. Based on review of the correspondence, VVB confirms that PP has informed ESKOM about updates regarding the project design at relevant stages. Further, VVB confirms that PP has designed an on-going stakeholder consultation mechanism to enable communication of grievances throughout the crediting period of the project activity.
--	--	--

	effect of other large scale renewable energy projects already feeding into South Africa's grid at the time of publishing the PDD was taken into account through the Build Margin (BM) calculation. The build margin refers to the group of prospective power plants whose construction and future operation would be affected by the proposed project activity. This was demonstrated to the VVB during the validation and met the requirements. Potential selling green power to a non-Eskom offtaker: This had been checked during the project design and was confirmed again when addressing these comments. No evidence was found to substantiate this, and in line with the point above on double counting it was also validated by the VVB. VVB has validated that the GHG emission reductions claimed by the project are based solely on the electricity produced and sold to Eskom in accordance with the PPA. Verification for emission reductions claimed is based on the calibrated meters measuring electricity that is fed to the national grid and there is no additional grid or feed that could be used for an additional offtake (wheeling) agreement or	7. Potential Legal dispute: PP has acknowledged the risk of legal disputes over the project lifetime. However, VVB did not find evidence of any ongoing legal dispute during the validation stage.
--	---	--

	direct feed to an independent offtaker. The sites are also remote and have little potential to feed adjacent plants or municipalities. Engagement with Eskom as a stakeholder: During the stakeholder identification process, Eskom had been identified as a key stakeholder and representatives of Eskom had been included on the stakeholder register and contacted via email. They were provided with stakeholder updates and feedback was requested. While it is acknowledged that Eskom is a large organization, the project proponent is open to engage with more than one Eskom representative or department should they be added to the stakeholder register. The stakeholder engagement process was validated and evidence and source documentation were verified by the independent VVB during the validation process. Potential for (non-public) legal disputes: It is acknowledged by the project proponents that legal disputes may arise with the project in the future for a number of reasons. While this comment is rather widely framed, it is acknowledged that	
--	--	--

	a number of risks existing in these (and most other projects) such as Force Majeure, Liability and Indemnity clauses, penalty clauses, third party obligations and changes to governing frameworks (including legislation, regulations, policy or the international voluntary carbon market requirements) or the amendment of contracts. Note the following acronyms are used: • REIPPPP: Renewable Energy Independent Power Producer Procurement Programme • RMI: Risk Mitigation Independent Power Producer Procurement Programme • PPA: Power Purchase Agreement (concluded between Eskom and an IPP) • IA: Implementation Agreement (concluded between the DMRE and/or its predecessor or successor government department and the IPP) • IPPO: Independent Power Producer Office (an office within the Department of Mineral Resources and Energy)	
--	--	--

	• DMRE: Department of Mineral Resources and Energy • IPP: Independent Power Producer (a project proponent)	
Comment 3 – Anonymous – submitted 20 May 2025 PUBLIC COMMENTARY PERIOD: PROJECT VCS 5488 Clarity is sought regarding the potential to generate carbon offsets from projects registered under the South African Renewable Energy Independent Power Producer Procurement Programme (REIPPPP). It is noted project Scatec RMI Hybrid Cluster (5488) , as well as projects De Aar Hydra Storage (5337) and Redstone CSP Project (5146) have been awarded contracts under this national renewable energy procurement programme. 1. CONTEXT The REIPPPP is a competitive tender programme established to attract private sector investment into grid-connected renewable energy generation in South Africa. The programme is playing a critical role in driving the country's transition to cleaner energy.	The response to the comment has been split into the primary issues raised as follows: • Clarity regarding potential to generate carbon credits • Management of carbon-related environmental attributes to avoid double counting • Environmental attributes already embedded in the GEF • If national utility (Eskom) acknowledges the rights to monetise carbon credits • Alignment with carbon accounting principles (related to GEF) • Financial structuring of REIPPPP tariffs within additionality • If PPA explicitly defines ownership	VVB has reviewed the public comment and assessed the following aspects of project concerning the comment: • The comment raises a concern that additionality of the project activity may be impacted due to its registration under the South African Renewable Energy Independent Power Producer Procurement Programme (REIPPPP). VVB has assessed the additionality of this project activity (5146) and confirms that the project activity is automatically additional as demonstrated in section 3.5.2 of the VCS PD. • The VVB has checked for double counting at the validation stage by reviewing all the relevant registries that could hold RECs from the project. The project activity is not

Additional context on the REIPPPP is provided in Box 1.	and transferability of the associated environmental rights	registered under any other emissions trading program or any other mechanism that includes GHG allowance trading. Following registries have been assessed:
Box 1: Overview of the REIPPPP and key stakeholders	Alignment with carbon accounting principles (related to REIPP projects already included in GEF)	➤ https://www.climateactionreserve.org/
The request aims to gain a deeper understanding of the eligibility of projects under the REIPPPP for the generation of carbon credits. Specifically, how environmental integrity is upheld and whether financial structures comply with international carbon market principles.	Clarity regarding potential to generate carbon credits: REIPPPP tariffs are structured for cost recovery and to ensure a reasonable return to investors. However, that does not preclude additionality in carbon market terms. Financial viability and additionality are not binary concepts — additionality is assessed in relation to whether the carbon revenue played a role in project decision-making or was necessary to overcome barriers (financial, technological, regulatory, etc.) at the time of investment.	➤ https://registry.goldstandard.org/projects?q=&page=1
Clarity is being sought on two principal aspects:	- The management of carbon-related environmental attributes to avoid double counting, and - The financial structuring of REIPPPP tariffs in relation to the principle of additionality.	➤ https://iceland.itmoregistry.net/Public/Project
2.1. Management of Environmental Attributes and Avoidance of Double Counting	Hence, not all REIPPPP projects are equally profitable or de-risked. Many face unexpected delays (e.g. grid constraints, currency issues, escalating capex, lead times in remote areas) or thin margins. These issues are material to an assessment of additionality.	➤ https://biocarbonregistry.com/en/projects/
It is understood that renewable energy generated through REIPPPP is included in the calculation of South Africa's national grid emission factor (GEF), which in turn is used to estimate the emissions attributable to grid-supplied electricity. The implication is that renewable electricity		➤ https://projects.globalcarboncouncil.com/pages/submitted_projects
		➤ https://registry.verra.org/app/search/VCS/All%20Projects
		➤ https://cdm.unfccc.int/Projects/projsearch.html
		➤ The project is not registered under the international REC Mechanism e.g. I-REC Device Registry and the same is confirmed through the I-REC web site (https://v-1.evident.app/Public/ReportDevices/).

supplied to the grid contributes to a lower overall GHG intensity, thereby benefiting all grid users.

There is therefore a concern that issuing carbon credits for the same environmental benefit, already embedded in a reduced GEF, could constitute a form of double counting. To support a credible carbon market approach, clarity is needed on how the environmental attributes are treated contractually and institutionally.

Specifically, the following aspects require investigation:

- Whether the Power Purchase Agreement (PPA) explicitly defines the ownership and transferability of the associated environmental rights.
- Whether Eskom, as the purchaser of electricity, acknowledges that the IPP retains the rights to monetise environmental attributes such as carbon credits.
- How the programme ensures alignment with carbon accounting principles, given that GHG reductions from REIPPPP projects are already reflected in the national GEF used for

So in principle these projects can qualify as carbon projects under the voluntary carbon market (as has already been demonstrated with a number of other REIPPPP projects in South Africa already registered under recognized carbon schemes such as Verra VCS or the CDM.

The generation and use of carbon offsets for South African domestic purposes – including under the Carbon Tax Act (15 of 2019) and its associated offset regulations – have been explicitly recognized and encouraged by the National Treasury and South African Revenue Services. This project is aligned with this objective and the resultant VCUs (carbon credits) from these projects qualify under the domestic offset scheme and it is expected that they will be used within this framework. They are intended to support domestic decarbonisation and net zero targets, and as such directly support South Africa's NDC and industrial transition.

Management of carbon related environmental attributes to avoid double counting: In accordance with

The project has provided all information required on double counting and double claiming in section 1.16 and 1.17 of the VCS PD. VVB has confirmed through interviews with representatives of PP that the project is not claiming and does not plan to claim GHG credits under any other GHG program. Further, PP has confirmed that the project activity has not been rejected under any other GHG program. A signed declaration stating the same is provided by PP to the VVB.

- VVB has reviewed the grid emission factor calculation demonstrated in Grid Emission factor calculation sheet/F2/. Based on the review, VVB confirms that the EF is calculated as per CDM TOOL 07. All the data sources referenced to determine the EF are found to be relevant, credible and latest. Therefore, VVB confirms that PP has determined the grid emission factor

greenhouse gas accounting by electricity consumers.

2.2. Alignment of Tariff Structures with the Principle of Additionality

The principle of additionality is foundational to the issuance of carbon credits. In the case of REIPPPP, awarded tariffs are designed to ensure financial viability and bankability. The Power Purchase Agreement allows for cost recovery, typically based on the plant's Levelised Cost of Electricity, which includes capital investment, the weighted average cost of capital, capacity factor, and both fixed and variable operating and maintenance costs.

Given this, there is a concern that the inclusion of carbon credit revenue could be considered surplus to financial requirements, thereby potentially undermining additionality. If projects are already financially viable based on tariff revenues structured to cover all costs and provide a reasonable return on investment, the addition of carbon credit income might be perceived as a windfall, rather than a necessity for project feasibility.

It would be useful to understand if the programme

the VCS Standard (§3.23), project proponents need to demonstrate that there is no double counting, or double selling of any GHG emission reductions. This was confirmed by cross referencing a number of local and international registries, including Renewable Energy Certificates (RECs). Further to this, during the project development process, each project owner provided a signed declaration that no other environmental benefits are sought, and no offtake agreements or GHG benefits have been allocated outside of the listed Verra project. These documents formed part of the independent validation by the VVB during the audit process. No additional contractual obligations or evidence of any carbon credits being claimed or accounted for by any other third party was found during this review.

In addition, during the project development process, each project owner provided a signed declaration that no other offtake agreements or GHG benefits have been allocated outside of the listed Verra project.

appropriately and sufficiently justified the sources.

- Based on the review of the PPA between the PP and ESKOM/B1/, and Implementation agreement between PP and The Department of Energy/B12/, VVB confirms that PP will sell the renewable energy generated through Redstone project activity only to ESKOM as per terms of agreement outlined in the PPA. Further, VVB based on on-site inspection confirms that there is no other grid present in proximity of project area at the time of validation site visit.
- VVB has reviewed the license issued by IPP/B2/ to the project proponent ACWA Power Solar Reserve Redstone Solar Thermal Power Plant (RF) (Pty) Ltd. Therefore, the legal ownership of the project activity is confirmed through review of the same. Additionally, VVB has reviewed an

has conducted any assessments or established guidelines to determine if carbon revenues are essential to the financial case for the project, or whether these are supplementary.

3. CONCLUSION

In closing, further insight into how the REIPPPP framework aligns with the requirements for carbon credit generation under international market standards would be appreciated. Specifically, if there are any applicable policy or regulatory provisions that may affect project eligibility, and whether certain bidding windows or project categories are treated differently. Clear guidance on these points will assist in evaluating the role of REIPPPP projects within credible carbon market mechanisms and support efforts to ensure environmental integrity and financial rigour in project development,

Environmental attributes already embedded in GEF: Under the VCM, South Africa does not have a published and/or approved national grid emission factor (GEF) or standardised baseline for the use by project proponents, and each project has to determine the Grid Emission Factor in accordance with the VCS Standard (§3.13) and the applied methodology. In the case of this project, CDM methodology ACM 0002 (v 22.0) was applied, and as per §5.2. The associated tool, CDM TOOL07 was applied in order to determine the GEF at the time of validation. This process is approved by Verra and takes cognisance of all components of the national grid, including renewable energy projects.

As per the prescribed tools, all non-operational projects, e.g. those seeking registration with Verra, as well as other REIPPPP projects not yet commissioned at the time of publishing the PDD are not included in the baseline scenario and hence are excluded from the grid emission factor calculation. However, the effect of all other large scale renewable energy projects already feeding into South Africa's

agreement between the PPs ACWA Power Solar Reserve Redstone Solar Thermal Power Plant (RF) (Pty) Ltd (Redstone') in whose name the IPP is issued and Anthesis B.V. ('Anthesis') who are involved in developing the project under VERRA. Based on this agreement, VVB confirms that both parties will have joint ownership of the carbon project when registered. Further, VVB has reviewed the legal opinion/B.11/ obtained by PP and confirms that there is no indication about exclusive ownership of carbon credits. Therefore, VVB confirms that PP has demonstrated ownership of the project activity and therefore carbon credits generated through the project activity in line with VCS standard.

grid at the time of publishing the PDD were taken into account through the Build Margin (BM) calculation. The build margin refers to the group of prospective power plants whose construction and future operation would be affected by the proposed project activity. This was demonstrated to the VVB during the validation and met the requirements.

If national utility (Eskom) acknowledges the rights to monetise carbon credits: The specific rights to monetise carbon credits does not directly form part of the VERRA registration process, however project proponents do need to demonstrate ownership of any VCUs. This was responded to in the comment above and were demonstrated to the VVB during the validation through legally binding agreements that the various project proponents had signed. These agreements covered commercial aspects of the project, including responsibilities for various project activities (such as operations, offtake, carbon development etc.).

While the comment suggests a contractual obligation

limiting the monetisation of the carbon credits to the project proponent, independent legal opinion and review obtained by both project proponents has a differing opinion in this, as the relevant clauses of both the PPA and the Implementation Agreements (IA) are ambiguous regarding the rights to monetise any environmental attributes. The environmental attributes are not clearly defined either in the PPA or IA. For example, for the Scatec RMI project, the PPA implicitly acknowledges the IPP's right to utilise carbon credits under the RMI. This implicit acknowledgement lies in the calculation of the IPP's cost recovery of carbon taxation payments made by the IPP on behalf of Eskom, which must be reduced by the maximum carbon offset permissible, whether the IPP actually earns any offsets, and thus implying that the value of such carbon offset is for the account of the IPP. Where a contract is ambiguous then one can rely on general principles of law to interpret the contract and, in this instance such reliance can be placed on the general principle of property law that the (carbon) project owner

(the IPP) is also owner of the environmental attributes with the right to monetise such environmental attributes. Legal review was conducted in the light of these comments and a letter to the IPP from the IPPO dated 1 July 2025. The review was unable to identify any substantive contractual foundation that differs from the IPP's current understanding and interpretation of the contracts and the eligibility to register a project and gain VCU from the project.

Furthermore, these contractual obligations also provide for changes to the operating environment in the future that may arise from changes in ownership, legal disputes, legislation, contractual amendments, policy changes and/or changes to the international voluntary carbon market, which the PP acknowledges may change any future claim or eligibility. These documents formed part of the independent validation by the VVB during the audit process .

In addition to the above, the monetization of carbon credits resultant from a VCS project, would fall outside of the registration and issuance

process, as this would be managed in terms of a commercial agreement between project proponents and shareholders regarding any revenue sharing.

Alignment with carbon accounting principles (related to GEF): Under the VCM, South Africa does not have a published and approved national grid emission factor or standardised baseline, and each project has to determine the Grid Emission Factor in accordance with the VCS Standard (§3.13) and the applied methodology. In the case of this project, CDM methodology ACM 0002 (v 22.0) was applied, and as per §5.2. CDM TOOL07 were applied in order to determine the grid emission factor. This process is approved by Verra and takes cognisance of all components of the national grid, including renewable energy projects.

As per the prescribed tools, the projects seeking registration with Verra, as well as other REIPPP projects not yet commissioned at the time of publishing the PDD are not included in the baseline scenario and hence are excluded from the grid emission factor calculation. However, the effect of other

large scale renewable energy projects already feeding into South Africa's grid at the time of publishing the PDD was taken into account through the Build Margin (BM) calculation. The build margin refers to the group of prospective power plants whose construction and future operation would be affected by the proposed project activity. This was demonstrated to the VVB during the validation and met the requirements.

Financial structuring of REIPPP tariffs within additionality: In accordance with the VCS Standard (§3.14) and the applied methodology (§5.2.3 and the associated Tools that were applied), additionality was evaluated within the framework applicable to the Verra VCS requirements, including CDM TOOL1 v7, CDM TOOL 24 v3.1 and CDM TOOL 27 v14. All parameters used in the assessment of additionality were valid at the time of investment decision, and were verified by the carbon project developer as well as the independent VVB before being submitted to Verra as part of the Registration process.

Even though the REIPPPP is a program implemented by the South African government, it is not a government subsidised program, but rather a price-competitive procurement programme. We acknowledge that REIPPPP tariffs are structured to provide cost recovery. However, the assumption that all awarded projects are fully bankable without carbon revenue is not borne out in practice. Of the projects commented on, only the Redstone CSP project is a REIPPPP project. The other projects were all granted licenses under the Risk Mitigation IPP Programme. Despite this subtle difference, the projects still had to demonstrate additionality in compliance with the Verra rules, irrespective if they were part of the REIPPPP programme or not.

Additionally, risks such as capex overruns, long development lead times, logistics issues at remote greenfield sites and currency (ZAR) devaluation and volatility impacted financial viability, reinforcing the importance of a carbon revenue stream.

The technology applied for the Redstone CSP is on a positive list, so additionality is automatic in terms of the current Verra guidelines. As such, the projects meet the criteria for additionality as per VCS rules.

If PPA explicitly defines ownership and transferability of the associated environmental rights: In accordance with the VCS Standard (§3.7), project proponents need to demonstrate the legal right to own, control and operate the project activities. This and the proof of right were demonstrated to the VVB during the validation through legally binding agreements between that the various project proponents had signed. These agreements covered commercial aspects of the project, including responsibilities for various project activities (such as operations, offtake, carbon development etc.).

While the comment suggests a contractual obligation limiting the ownership and transferability of the carbon credits to the project proponent, independent legal opinion and review obtained

by both project proponents has a differing opinion in this, as the relevant clauses of both the PPA and the Implementation Agreements (IA) are ambiguous regarding the ownership. Where a contract is ambiguous then one can rely on general principles of law to interpret the contract, and in this instance reliance can be placed on the general principle of property law that the (carbon) project owner (the IPP) is also owner of the environmental attributes with the right to monetise such environmental attributes. Legal review was conducted in the light of these comments and a letter to the IPP from the IPPO dated 1 July 2025. The review was unable to identify any substantive contractual foundation that differs from the IPP current understanding and interpretation of the contracts and the eligibility to register a project and gain VCUs from the project.

Furthermore, these contractual obligations also made provision for changes to the operating environment in the future that may arise from changes in ownership, legal disputes, legislation, contractual amendments, policy changes and/or

changes to the international voluntary carbon market, which the PP acknowledges may change any future claim or eligibility. These documents formed part of the independent validation by the VVB during the audit process .

Alignment with carbon accounting principles (related to REIPP projects already included in GEF): Under the VCM, South Africa does not have a published and approved national grid emission factor or standardised baseline, and each project has to determine the Grid Emission Factor in accordance with the VCS Standard (§3.13) and the applied methodology. In the case of this project, CDM methodology ACM 0002 (v 22.0) was applied, and as per §5.2. CDM TOOL07 were applied in order to determine the grid emission factor. This process is approved by Verra and takes cognisance of all components of the national grid, including renewable energy projects.

As per the prescribed tools, the projects seeking registration with Verra, as well as other REIPPP projects not yet commissioned at the

time of publishing the PDD are not included in the baseline scenario and hence are excluded from the grid emission factor calculation. However, the effect of other large scale renewable energy projects already feeding into South Africa's grid at the time of publishing the PDD was taken into account through the Build Margin (BM) calculation. The build margin refers to the group of prospective power plants whose construction and future operation would be affected by the proposed project activity. This was demonstrated to the VVB during the validation and met the requirements.		
Comment 4 – Independent Power Producer (IPP) Office - Elsa Strydom – submitted 10 June 2025 Dear Verra, FORMAL CONCERN REGARDING REGISTRATION OF IPP PROJECTS UNDER THE VERRA VCS PROGRAMME	The response to the comment has been split into the primary issues raised as follows: - Carbon credits are subject to contractual provisions (§4) - IPP prohibited from registering Environmental Attributes unless prior consent obtained (§5)	VVB has reviewed the public comment and assessed the following aspects of project concerning the comment: VVB has reviewed the license issued by IPP/B2/ to the project proponent ACWA Power Solar Reserve Redstone Solar Thermal Power Plant (RF) (Pty) Ltd. Therefore, the legal ownership of the project activity is confirmed through

1. We refer to the above-subject matter. 2. It has come to our attention that the following Independent Power Producer (IPPs) Projects have been registered or started the process of registering under the Verra Verified Carbon Standard (VCS) programme: 2.1 Scatec RMI Hybrid Cluster (VCS 5488) 2.2 De Aar Hydra Storage (VCS 5337) 2.3 Oya Hybrid Project (VCS 5312) 2.4 Umoyilanga Hybrid Project (VCS 5272) 2.5 Redstone CSP Project (VCS 5146) 3. These projects are being implemented under the Risk Mitigation Independent Power Producer Procurement Programme (RMIPPPP) and Renewable Energy Independent Power Producer Procurement Programme (REIPPPP), managed by the Department of Electricity and Energy through the IPP Office. 4. We wish to formally inform Verra that, in accordance with the Power Purchase Agreements (PPAs), Implementation Agreements	• Prior authorization required otherwise may be inconsistent with contracts (§6) • Verra to review double counting (including contractual obligations) (§7) • Department request to meet with Verra (§8) Carbon credits are subject to contractual provisions (§4): Independent legal review by the project proponents came to the conclusion that this is a factually incorrect statement because there are no specific contractual provisions relating to carbon credits and/or other claims to environmental attributes under any of the REIPPPP and/or RMI contractual and other documentation to which this comment refers. In addition to the above, the monetization of carbon credits resultant from any VCS project, would fall outside of the registration and issuance process, as this would be a commercial agreement between project proponents and shareholders regarding any	review of the same. Additionally, VVB has reviewed an agreement between the PPs ACWA Power Solar Reserve Redstone Solar Thermal Power Plant (RF) (Pty) Ltd (Redstone') in whose name the IPP is issued and Anthesis B.V. ('Anthesis') who are involved in developing the project under VERRA. Based on this agreement, VVB confirms that both parties will have joint ownership of the carbon project when registered. Further, VVB has reviewed the legal opinion obtained by PP/B.11/ and confirms that there is no indication about either exclusive ownership of carbon credits or a requirement to obtain prior consent from REIPPPP or RMI or ESKOM. Therefore, VVB confirms that PP has demonstrated ownership of the project activity and therefore, carbon credits generated
---	---	--

(IAs), and associated procurement documents governing these programmes, the ownership and treatment of environmental attributes, including carbon credits and associated claims, are subject to specific contractual provisions. 5. Under these provisions, IPPs are prohibited from registering, monetizing, or otherwise utilizing Environmental Attributes outside of the approved frameworks, unless prior consent has been obtained from the Department. The Department retains the sole discretion to determine the use, or monetization of Environmental Attributes in terms of the relevant environmental policies. 6. Based on our current understanding, unless prior written authorisation has been obtained from the Department, the IPPs do not hold the exclusive rights to register such environmental attributes under voluntary carbon markets, including the VCS. Accordingly, such registrations, if made without the necessary approval, may be inconsistent with the contractual obligations under which these projects were awarded and developed.	revenue sharing. The VVB reviewed current contracts between the PPs regarding carbon credits as part of the validation process. IPP prohibited from registering Environmental Attributes unless prior consent obtained (§5): Independent legal review by the project proponents came to the conclusion that this is a factually incorrect statement because there are no specific contractual provisions relating to carbon credits and/or other claims to environmental attributes under any of the REIPPPP and/or RMI contractual and other documentation to which this comment refers. Given this lack of contractual specificity, it was deemed that: • IPPs are not prohibited from registering, monetizing, or otherwise utilizing Environmental Attributes outside of the approved frameworks, • the Department does not retain any discretion to determine the use, or monetization of environmental attributes, and	through the project activity in line with VCS standard. • The VVB has checked for double counting at the validation stage by reviewing all the relevant registries that could hold RECs from the project. The project activity is not registered under any other emissions trading program or any other mechanism that includes GHG allowance trading. Following registries have been assessed: ➤ https://www.climateactionreserve.org/ ➤ https://registry.goldsstandard.org/projects?q=&page=1 ➤ https://iceland.itmoregistry.net/Public/Project ➤ https://biocarbonregistry.com/en/projects/ ➤ https://projects.globalcarboncouncil.com/pages/submitted_projects ➤ https://registry.terra.org/app/search/VCS/All%20Projects ➤ https://cdm.unfccc.int/Projects/projsearch.html
---	---	---

7. Given the potential implications of double-counting, the Department requests that Verra reviews the registration and claims associated with the listed projects, considering the contractual limitations and rights under the REIPPPP and RMIPPPP. 8. We would also like to arrange an engagement directly with Verra on issues related to environmental attributes and provide clarifications regarding the relevant contractual and procurement frameworks. 9. For further information and to confirm your availability for such an engagement, please contact Ms. Lena Mangondo, Head of Legal at the IPP Office, at lena.mangondo@ipp-projects.co.za 10. We look forward to your response. Yours sincerely, ELSA STRYDOM INTERIM HEAD: IPP OFFICE DATE: 2025/06/10	- there are no environmental policies that specifically address these issues except for regulations under the Carbon Tax Act which provide for a process in terms of which project proponents (including project proponents of REIPPPP projects) can seek authorisation to list carbon credits as carbon offsets on the Carbon Offset Administration System. proponent. Prior authorization required otherwise may be inconsistent with contracts (§6): Independent legal review by the project proponents came to the conclusion that this is a factually incorrect statement because there are no specific contractual provisions relating to carbon credits and/or other claims to environmental attributes under any of the REIPPPP and/or RMI contractual and other documentation to which this comment refers. Given this lack of contractual specificity: - IPPs do not need prior authorisation from the Department to register projects under the VCS,	➤ The project is not registered under the international REC Mechanism e.g. I-REC Device Registry and the same is confirmed through the I-REC web site (https://v-1.evident.app/Public/ReportDevices/). The project has provided all information required on double counting and double claiming in section 1.16 and 1.17 of the VCS PD. VVB has confirmed through interviews with representatives of PP that the project is not claiming and does not plan to claim GHG credits under any other GHG program. Further, PP has confirmed that the project activity has not been rejected under any other GHG program. A signed declaration stating the same is provided by PP to the VVB.
---	---	---

indeed a number of REIPPP projects have been registered and operational for some years, including having transacted VCU into the carbon. Clearly, therefore no prior authorisation is required.

Verra to review double counting (including contractual obligations) (§7):

In accordance with the VCS Standard (§3.23), project proponents need to demonstrate that there is no double counting, or double selling of any GHG emission reductions. This was confirmed by cross referencing a number of local and international registries, including Renewable Energy Certificates (RECs). Further to this, during the project development process, each project owned provided a signed declaration that no other environmental benefits are sought offtake agreements or GHG benefits have been allocated outside of the listed Verra project. These documents formed part of the independent validation by the VVB during the audit process.

In addition, during the project development

	process, each project owned provided a signed declaration that no other offtake agreements or GHG benefits have been allocated outside of the listed Verra project. Department request to meet with Verra (§8): The Project proponent acknowledges that a meeting is requested and no further action is required currently.	
--	--	--

3.2.2 Risks to Local Stakeholders and the Environment

3.2.2.1 Management Experience

VVB has interviewed representatives of Anthesis and confirms that the management has extensive experience in voluntary carbon market.

VVB has interviewed representatives of PP to confirm that the ACWA Power management is experienced in construction and operation of similar projects. VVB confirms that PP, ACWA Power has experience is implementation of following projects of similar nature :

- Bokpoort, South Africa (<https://acwapower.com/en/projects/bokpoort-csp-ipp/>)
- Noor Energy, UAE (<https://acwapower.com/en/projects/noor-energy-1/>)
- Noor I (<https://acwapower.com/en/projects/nooro-i-csp-ipp/>) and Noor II (<https://acwapower.com/en/projects/nooro-ii-csp-ipp/>) in Morocco

3.2.2.2 Risk Assessment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Natural and human-induced risks to stakeholders' wellbeing	PP has identified potential risks and described the same in section 2.2.2 of the PD along with their mitigation measures. VVB has validated that impact due to construction activities will be mitigated by following an ecological approach towards biodiversity.

	A perimeter fence will be erected to minimize dust formation. This was verified during the on-site audit. Light pollution will be mitigated by maintaining heliostats and reducing spillage beyond project sites. Noise pollution will be mitigated during operation phase as specified for industrial districts in SANS 10103. Visual impact will be minimized by keeping vegetation removal to possible. Additionally, VVB without limiting itself to the information provided by PP has inspected additional potential risks due to the activity by interviewing the local stakeholders to check if there existed any concern regarding the project design as construction of tower was complete at the time on on-site audit. However, no such concern was observed. Based on above information, VVB concludes that the measures taken by PP to reduce risk to stakeholders well-being are sufficient.
Risks to stakeholder participation	Based on review of the MR, interviews with representatives of PP and the end users, the verification team confirms that no risks were identified related to stakeholders' participation.
Working conditions	Based on review of the MR, interviews with representatives of PP and employees of the PP , the verification team confirms that no risks were identified related to working conditions. PP has specified that the project activity will comply with local legislation, specifically the Occupational Health and Safety Act (Act No 85 of 1993) and Basic Conditions of Employment Act (No. 75 of 1997).
Safety of women and girls	Based on review of Redstone's Code of Conduct and Ethics Policy, interviews with representatives of PP, VVB confirms that no risks were identified related to safety of women and girls.
Safety of minority and marginalized groups, including children	Based on review of Redstone's Code of Conduct and Ethics Policy, interviews with representatives of PP, VVB confirms that no risks were identified related to safety of women and girls.

Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials, and chemical pesticides and fertilizers)	PP has specified that industrial wastewater to be generated by the project is classified as a moderate hazard, with a Hazard rating of 3. VVB has reviewed the test report to confirm the same. VVB has reviewed SABS 0228 code to confirm that brine is an inorganic process waste or residue and was classified as class 6 (Poisonous (toxic) substances). Accordingly, PP has specified that the ponds will therefore be lined with a triple liner and double drainage system as required by the Department of Water and Sanitation (DWS). Based on review of the mitigation measure taken, VVB confirms that, the risk will be sufficiently mitigated.
---	---

3.2.3 Respect for Human Rights and Equity

3.2.3.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination	Based on review of Redstone's Code of Conduct and Ethics Policy/B6/, interviews with representatives of PP, VVB confirms that no risks were identified related to discrimination at workplace.
Sexual harassment	Based on review of Redstone's Code of Conduct and Ethics Policy, interviews with representatives of PP, VVB confirms that no risks were identified related to sexual harassment at workplace.
Equal pay for equal work	Based on review of Redstone's Code of Conduct and Ethics Policy, interviews with representatives of PP, VVB confirms that no risks were identified related to equal pay for equal work.
Gender equity in labor and work	Based on review of Redstone's Code of Conduct and Ethics Policy, interviews with representatives of PP, VVB confirms that no risks were identified related to gender equity in labor and work.
Forced labor	Based on review of Redstone's Code of Conduct and Ethics Policy, interviews with representatives of PP, VVB confirms that no risks were identified related to forced labor.
Child labor	Based on review of Redstone's Code of Conduct and Ethics Policy, interviews with representatives of PP, VVB confirms that no risks were identified related to child labor.

Human trafficking	Based on review of Redstone's Code of Conduct and Ethics Policy, interviews with representatives of PP, VVB confirms that no risks were identified related to human trafficking.
-------------------	--

3.2.3.2 Human Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risks identified	Based on review of the EIA report of the project activity, VVB confirms that Negative impacts were considered negligible and mitigated as far as possible.

3.2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
Destruction of archaeological sites because of construction activities	The risk will be mitigated by alerting the SAHRA APM unit in case of discovery of archaeological site or remains. VVB concludes that the measures taken by PP to reduce risk to cultural heritage are sufficient.
Destruction of cemeteries or graves because of construction activities.	PP has specified that, The site layout has been adjusted to avoid cemeteries and the cemeteries are fenced with a 10-meter buffer. VVB concludes that the measures taken by PP to avoid destruction of cemeteries or graves are sufficient.

3.2.3.4 Property Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risks identified	VVB has confirmed through review of land lease agreement that the owner has consented to the project activity being constructed and operational on the property. Additionally, VVB has interviewed the local stakeholders particularly the Mayor of the municipality under which the project is implemented to confirm that the project activity does not impact customary rights of Indigenous people and local communities.

3.2.3.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Process used to design the benefit sharing plan	The project activity is constructed on private land as confirmed through land-lease agreement. Therefore, a benefit sharing mechanism is not required.
Summary of the benefit sharing plan	The project activity is constructed on private land as confirmed through land-lease agreement. Therefore, a benefit sharing mechanism is not required.
Approval and dissemination of benefit sharing plan	The project activity is constructed on private land as confirmed through land-lease agreement. Therefore, a benefit sharing mechanism is not required.

3.2.4 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Impacts on biodiversity and ecosystems	PP has specified that, as per approved EMPr the risk to indigenous natural vegetation will be mitigated by rehabilitation as soon as construction is complete. A faunal register must be maintained to record avifaunal injuries and fatalities. VVB has reviewed the EMPr and confirms that sufficient mechanism is in place to mitigate the risk.
Soil degradation and soil erosion	No risks were identified, and the implementation will be as per the EMPr.
Water consumption and stress	No risks were identified, and the implementation will be as per the EMPr.

3.2.4.1 Rare, Threatened, and Endangered species

Risk identified	Evidence gathering activities, evidence checked, and assessment conclusion
Habitats for rare, threatened, and endangered species	A total of 56 Red Data animals (excluding avifauna) are known to occur in the Northern Cape Province. Of these species, 12 are listed as Data Deficient (DD), 21 are listed as Near Threatened (NT), 12 are listed as Vulnerable (VU), 5 are listed as Endangered (EN) and 5 species are listed as Critically Endangered (CR). It is estimated that approximately 79% of these species have a low probability of occurring in the immediate region of the project area. 11 species are estimated to have a moderate probability of occurring and two species have a high probability of occurrence in the immediate region of the project area. South African Bird Atlas Project (SABAP) records 168 bird species of which 11 are Red Listed Species. The presence of red data species in the project area was assessed based on the avifaunal impact assessment specialist study carried out as part of the EIA process. VVB has reviewed the EIA assessment to confirm the same. The implementation will be as per the EMP to mitigate the risks to species and habitat of rare, threatened and endangered species.
Areas needed for habitat connectivity	Impacts from the proposed development are unlikely to increase regional or local levels of fragmentation and habitat isolation significantly. VVB has reviewed the EIA report to confirm the same.

3.2.4.2 Introduction of Species

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
No invasive species will be used as part of project activity.	Not applicable

3.2.4.3 Ecosystem conversion

Risks Identified	Evidence gathering activities and evidence checked
Ecosystem conversion	Not applicable as this is not an ARR, ALM, WRC or ACoGS project.

3.3 Application of Methodology

3.3.1 Title and Reference

Type (methodology, tool or module).	Reference ID,	Title	Version
Methodology	ACM0002	Grid-connected electricity generation from renewable sources	22.0
Tool	TOOL01	Tool for the demonstration and assessment of additionality	7.0
Tool	TOOL07	Tool to calculate the emission factor for an electricity system	7.0
Tool	TOOL32	Positive lists of technologies	4.0

3.3.2 Applicability

Methodology ID	Applicability condition	Assessment and conclusion
ACM0002 version 22.0	This methodology is applicable to grid connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s);	The project activity involves installation of a greenfield power plant (a) that is connected to the national grid of South Africa. This is confirmed through power purchase agreement signed between PP and ESKOM/B1/. Thus, the applicability criteria is met by the project activity.

	(d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s); or Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid.	
...	In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic¹ or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s);	This project activity does not include the integration of a BESS. This was confirmed through review of the project technology/C1/, interviews with representatives of PP and on-site inspection.

	(d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s); (e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is located at site of the greenfield renewable power plant.	
	The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit must have started commercial operation prior to the start of a minimum historical reference period of five years. The reference period is used for the calculation of baseline emissions and defined in the	The proposed project activity is concentrated solar power plant, and the conditions are not applicable.

baseline emission section. Furthermore, no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity;

(c) In case of Greenfield project activities applicable under paragraph 7(a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g., by referring to feasibility studies or investment decision documents);

The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies² may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more

	than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period. (e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in coordination with the PSP during pumping mode.	
	In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using	The proposed project activity is concentrated solar power plant, and the conditions are not applicable.

equation (7), is greater than 4 W/m²; or

(c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or

(d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², and all of the following conditions shall apply:

(i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²;

(ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity;

(iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² are:

a. Lower than or equal to 15 MW; and

b. Less than 10 per cent of the total installed

	capacity of integrated hydro power project.	
	In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the	The proposed project activity is concentrated solar power plant, and the conditions are not applicable.

	implementation of the CDM project activity.	
	In the case of PSP, the project participants shall demonstrate in the VCS Project Description that the project is not using water which would have been used to generate electricity in the baseline.	The proposed project activity is concentrated solar power plant, and the conditions are not applicable.
	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	The proposed project activity is a greenfield concentrated solar power plant, and the conditions are not applicable.
	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking	The proposed project activity is a greenfield concentrated solar power plant, and the conditions are not applicable.

	business as usual maintenance”.	
CDM TOOL 01	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	The applicability criteria is met as application of tool is mandatory under the applied methodology.
	Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	As per the applied methodology, application of the tool is mandatory. Thus, the applicability criteria is met by the project activity.
CDM TOOL 07	This tool may be applied to estimate the Operating Margin (OM), Build Margin (BM) and/or Combined Margin (CM) when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	The project activity involves installation of a greenfield power plant (a) that is connected to the national grid of South Africa. This is confirmed through power purchase agreement signed between PP and ESKOM. Thus, the applicability criteria is met by the project activity.
	Under this tool, the emission factor for the project	Based on review of the project design, VVB confirms that the

	electricity system can be calculated either for grid power plants only or, as an option, can include offgrid power plants. In the latter case, two sub-options under step 2 of the tool are available to the project participants, i.e. option II a and option IIb. If option II a is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	project activity does not include off-grid power plants. Thus, the applicability criteria is not applicable to the project activity.
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity involves installation of a greenfield power plant (a) that is connected to the national grid of South Africa which is a non-Annex I country. This is confirmed through power

		purchase agreement signed between PP and ESKOM. Thus, the applicability criteria is met by the project activity.
	Under this tool, the value applied to the CO2 emission factor of biofuels is zero.	Based on review of the project design, VVB confirms that the project activity does not involve biofuels. Thus, the applicability criteria is not applicable to the project activity.
CDM TOOL 32	The use of this methodological tool is not mandatory for the project participants of a CDM project activity or CDM PoA for demonstrating their additionality.	The applicability criteria is met.
	This methodological tool shall be applied in conjunction with a small-scale or large-scale methodology which refers to this tool.	Based on review of the project design, VVB confirms that the project applied large-scale ACM0002 methodology. Thus, the applicability criteria is met by the project activity.
	The positive lists as contained in section 5 of this tool are valid up to 10 March 2025. Notwithstanding the provisions on the validity of new, revised and previous versions of methodologies and methodological tools in the “Procedure: Development, revision and clarification of baseline and monitoring methodologies	The tool is valid during the onset of this validation. Thus, the applicability criteria is met by the project activity.

	and methodological tools”, there will be no grace period for the application of this tool and the validity of the positive list after this date, including in cases where further technologies are added to the positive list through revisions of this tool before this date.	
--	--	--

3.3.3 Project Boundary

The project boundary and identified GHG sources, sinks and reservoirs for the project and baseline scenarios (including leakage if applicable) are appropriately defined in the VCS PD /A1/. The selection and justification for inclusion or exclusion is appropriate and duly supported by the observation during the on-site site visit. In addition to the table, a diagram of the project boundary, showing the physical locations of the various installations as part of the project activity are included in the PD. The choice of GHGs is also appropriate to the context of the project description. There is no GHG source that is omitted.

The relevant GHG sources included in or excluded from the project boundary are shown on the Table below:

Source		Gas	Included?	Assessment and conclusion
Baseline	Grid Connected Electricity Generation	CO ₂	Yes	Main emission source.
		CH ₄	No	Excluded for simplification. Minor source of emissions. Conservative
		N ₂ O	No	Excluded for simplification. Minor source of emissions. Conservative
		Other	No	No other GHG emissions relevant
Project	Concentrated Solar Plant Project Activity	CO ₂	No	Fossil fuels only used for emergency back-up generator.
		CH ₄	No	Not accounted as no or very little CH ₄ emitted by the project activity
		N ₂ O	No	Not accounted as no or very little N ₂ O emitted by the project activity
		Other	No	No other GHG emissions

3.3.4 Baseline Scenario

The project activity involves the installation of a newly built and grid-connected renewable power plant (i.e. Concentrated solar plant) and supplying the generated electricity to South African national grid, hence, according to the applied methodology ACM0002, version 22.0, the baseline scenario is determined properly as: “If the project activity is the installation of a Greenfield power plant with or without Battery Energy Storage System (BESS), the baseline scenario is electricity delivered to the grid by the project activity that would otherwise have been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in AM TOOL07 ‘Tool to calculate the emission factor for an electricity system (Version 07.0)’.”

The latest available version for “Tool to calculate the emission factor for an electricity system” is version 07.0 /A12/ and the PP has correctly referred to the same in the section 3.1 of the final VCS PD /1/ for determining baseline grid emission factor. PP has estimated the South African Grid Emission Factor based on the historical data published by ESKOM. The baseline emission factor for the electricity system has been calculated on ex-ante basis and will remain fixed for the entire project crediting period.

Hence, the baseline determined in the VCS PD is correct and in line with the applied methodology ACM0002, version 22.0. Also, the identified baseline for the project activity is the most likely scenario of what would have occurred in the absence of the project activity and is confirmed by validation team; thus, it is accepted. The project participant has included all sources and references used for baseline determination for the project activity in the PD/1/ and the identified baseline is justified appropriately by the project participant. The Baseline scenario and baseline emission calculations are found as per ACM0002 version 22.0/4/. The combined margin approach is the ex-ante approach as per tool to calculate the emission factor for an electricity system. Based on the above assessment, VVB confirms that:

- All the assumptions and data used by the PP are listed in the VCS PD, including their references and sources;
- All documentation used is relevant for establishing the baseline scenario and correctly quoted and interpreted in the PD;
- Assumptions and data used in the identification of the baseline scenario are justified appropriately, supported by evidence and can be deemed reasonable;
- Relevant national and/or sectoral policies and circumstances are considered and listed in the PD;
- The approved baseline methodology has been correctly applied to identify the most reasonable baseline scenario and the identified baseline scenario reasonably represents what would occur in the absence of the project activity.

3.3.5 Additionality

As per paragraph 13 §5.2.1 (a) of CDM TOOL 32 (Positive lists of technologies v 04.0). Solar thermal electricity generation including concentrating solar power appears on the positive list. It is therefore deemed additional if, at the time of the VCS Project Description submission, the percentage share of total installed capacity of the specific technology in the total installed grid connected power generation capacity in the host country is equal to or less than two per cent.

At the time of project submission, the total installed grid connected power generation capacity in South Africa was 53,898 MW. The total installed CSP capacity in South Africa was 600 MW, which is 1.1% of the total installed capacity which was verified through review of IPP project database (<https://www.ipp-projects.co.za/ProjectDatabase>) The project is therefore deemed automatically additional.

3.3.6 Quantification of GHG Emission Reductions and Carbon Dioxide Removals

Baseline Emission Calculation

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EGP_{J,y} \times EF_{grid,CM,y} \quad \text{.....Equation 1 (ACM0002 Equation 11)}$$

Where:

BE_y = Baseline emissions in year y (tCO₂/yr)

$EGP_{J,y}$ = (MWh/yr) Net electricity fed into the grid supplied by the project activity in year y

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using TOOL07 (t CO₂/MWh)

The results obtained are: 478,316.48 tCO₂/ annum, as shown below:

Parameter name	Value	Unit
BE_y	478, 316.48	t CO ₂ / yr
$EGP_{J,y}$	500,000.00	MWh/yr
$EF_{grid,CM,y}$	0.96	t CO ₂ /MWh

The project is a greenfield power plant and therefore $EG_{PJ,y}$ is calculated as follows:

$$EG_{PJ,y} = EG_{facility,y} \quad \text{.....Equation 2 (ACM0002 Equation 12)}$$

Where:

$EG_{PJ,y}$ = Net electricity generation fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)

$EG_{facility,y}$ = Net electricity generation supplied by the project plant to the grid in year y (MWh/yr)

The results obtained are: 500,000 MWh / yr.

Grid emission factor calculation

The grid emission factor is calculated as per TOOL 07 version 07 which includes following steps:

1. Step 1: Identify the relevant electricity systems

PP has identified the electricity system as grid of South Africa. This is deemed acceptable to the VVB.

2. Step 2 : Choose whether to include off-grid power plants in the project electricity system (optional)

The step is optional, and PP has chosen not to include the off-grid power plants in the project electricity system. This is deemed acceptable.

3. Step 3 : Select a method to determine the operating margin (OM)

PP has demonstrated that the share of electricity supplied by Low-Cost/Must-Run (LCMR) plants in the last 5 years has ranged from 19% – 23% and is consistently lower than 50%. Therefore, PP has selected simple OM method to calculate OM. This is deemed acceptable to the VVB.

4. Step 4: Operating margin calculation

VVB has cross-checked that calculation and confirms that the operating margin is calculated in accordance with TOOL 07 version 07 and the OM is determined as 1.00 tCO₂/MWh.

5. Step 5: Calculate the build margin (BM) emission factor

VVB has cross-checked that calculation and confirms that the build margin is calculated in accordance with TOOL 07 version 07 and the BM is determined as 0.81 tCO₂/MWh.

6. Step 6: Calculate the combined margin (CM) emission factor

VVB has cross-checked that calculation and confirms that the combined margin is calculated in accordance with TOOL 7 version 07 and the CM is determined as 0.96 tCO₂/MWh.

VVB has reviewed the grid emission factor calculation and confirms that the stepwise calculation as demonstrated in PD is in line with TOOL 07 requirements.

Project emissions calculation:

As per the applied methodology, the project emissions are estimated to be zero, since the project activity will use fossil fuel generator only as backup. The plant's auxiliary electricity consumption is approximately 25,000 MWh per year. However, the project activity involves net metering. Therefore, auxiliary consumption of the project activity will not be reflected in the final electricity generated values. This has been confirmed by VVB through interviews with representatives of PP and on-site inspection.

Leakage emissions calculation:

As per the applied methodology, the leakage emissions are estimated to be zero.

3.3.7 Methodology Deviations

No methodology deviations are identified.

3.3.8 Monitoring Plan

The project activity has correctly applied the monitoring methodology ACM0002, version 22.0 titled "Grid-connected electricity generation from renewable sources". The monitoring plan provides detailed information related to the collection and archiving of all relevant data needed to:

- Estimate or measure emissions occurring from GHG sources, sinks and reservoirs
- Determine the baseline emissions

The monitoring plan as per ACM0002 has been clearly described in section 5 of the VCS PD. It covers all the monitoring parameters required to monitor by the project activity and emission reductions due to the project activity accurately.

The monitoring plan/procedure followed to measure the emission reduction is applied accurately and with a conservative approach.

Parameters Determined ex-ante

The following parameters are determined ex-ante and mentioned in section 5.1 of the PD:

Parameter	Unit	Value	Assessment
EF _{grid,OM,y}	tCO ₂ /MWh	1.00	PP has determined the value using Equation 1, as per TOOL07 “Tool to calculate the emission factor for an electricity system”.
EF _{grid,BM,y}	tCO ₂ /MWh	0.82	PP has determined the value using Equation 6, as per TOOL07 “Tool to calculate the emission factor for an electricity system”.
EF _{grid,CM,y}	tCO ₂ /MWh	0.96	PP has determined the value using Equation 7, as per TOOL07 “Tool to calculate the emission factor for an electricity system”.

ShareLCMR	%	<table><tr><td>2021</td><td>2022</td><td>2023</td></tr><tr><td>19.4%</td><td>21.3%</td><td>23.1%</td></tr></table>			2021	2022	2023	19.4%	21.3%	23.1%	The values are determined from electricity generation data reported in Eskom's annual Integrated Report for 2023, and records published on Eskom's website for each power station.
		2021	2022	2023							
19.4%	21.3%	23.1%									
EGLCMR,y	MWh	<table><tr><td>EGLCMRy: 2023</td><td>49,691,000</td></tr><tr><td>EGLCMRy: 2022</td><td>48.092,000</td></tr><tr><td>EGLCMRy: 2021</td><td>42,495,000</td></tr></table>		EGLCMRy: 2023	49,691,000	EGLCMRy: 2022	48.092,000	EGLCMRy: 2021	42,495,000	The values are determined from electricity generation data reported in Eskom's annual Integrated Report for 2023, and records published on Eskom's website for each power station.	
		EGLCMRy: 2023	49,691,000								
		EGLCMRy: 2022	48.092,000								
EGLCMRy: 2021	42,495,000										
totaly	MWh	<table><tr><td>totaly: 2023</td><td>215,318,000</td></tr><tr><td>totaly: 2022</td><td>226,226,000</td></tr><tr><td>totaly: 2021</td><td>219,423,000</td></tr></table>		totaly: 2023	215,318,000	totaly: 2022	226,226,000	totaly: 2021	219,423,000	The values are determined from electricity generation data reported in Eskom's annual	
		totaly: 2023	215,318,000								
		totaly: 2022	226,226,000								
totaly: 2021	219,423,000										

			Integrated Report for 2023, and records published on Eskom's website for each power station.
Y	year	2023	This is the most recent year for which data is available at the time of validation.
<i>EF_{grid,OMsimple,y}</i>	tCO ₂ e/MWh	1.00	PP has determined the value using Equation 1, as per TOOL07 "Tool to calculate the emission factor for an electricity system".
<i>EG_{m,y}</i>	MWh	171,130,842	The values are determined from electricity generation data reported in Eskom's annual Integrated Report for 2023, and records published on Eskom's website for

			each power station.
<i>AEGtotal</i>	GWh	184,165,63	The values are determined from electricity generation data reported in Eskom's annual Integrated Report for 2023, and records published on Eskom's website for each power station.
<i>SET5 units</i>	Dimensionless	Kusile (2017/08/30) Medupi (2015/08/22) Sere Wind (2015/03/01) Komati (2009/03/31) Grootvlei (2008/04/40)	The values are determined from electricity generation data reported in Eskom's annual Integrated Report for 2023, and records published on Eskom's website for each power station.
<i>AEGset-5-units</i>	GWh	35,365.31	The values are determined from electricity generation data reported

			in Eskom's annual Integrated Report for 2023, and records published on Eskom's website for each power station.
$SET \geq 20\%$	Dimensionless	Kusile (2017/08/30) Medupi (2015/08/22) Ankerlig (2007/03/01) Gourikwa (2007/07/01) Grootvlei (2008/04/30) Komati (2009/03/31) Sere Wind (2015/03/01)	The values are determined from electricity generation data reported in Eskom's annual Integrated Report for 2023, and records published on Eskom's website for each power station.
$AEG_{set} \geq 20\%$	GWh	38,379.02	The values are determined from electricity generation data reported in Eskom's annual Integrated Report for 2023, and records published on Eskom's website for

			each power station.
<i>SETsample</i>	Dimensionless	Kusile (2017/08/30) Medupi (2015/08/22) Sere Wind (2015/03/01)	The values are determined from electricity generation data reported in Eskom's annual Integrated Report for 2023, and records published on Eskom's website for each power station.
<i>AEGset sample</i>	GWh	32,579,69	The values are determined from electricity generation data reported in Eskom's annual Integrated Report for 2023, and records published on Eskom's website for each power station.
<i>SETsample-CDM</i>	Dimensionless	Shown in Table 14 of the PD.	The values are determined from electricity generation data reported

			in Eskom's annual Integrated Report for 2023, and records published on Eskom's website for each power station.
<i>AEGSET-sample-CDM</i>	GWH	36,861.54	The values are determined from electricity generation data reported in Eskom's annual Integrated Report for 2023, and records published on Eskom's website for each power station.
<i>wOM</i>	%	75	The weighting of operating margin emissions factor is determined as per Tool07 paragraph 86 (a)
<i>wBM</i>	%	25	The weighting of build margin emissions factor is determined as per Tool07

			paragraph 86 (a)
BEy	tCO2e/yr	478,316	The value is estimated using equation 1 of the applied methodology.

Parameters Monitored ex-post

Monitoring of the project activity involves all the parameters necessary for calculation of GHG emission reduction by the proposed project activity. These parameters are mentioned in section 5.2 of the PD. The parameters, which are to be monitored include:

Parameter	Unit	Value		Assessment
EGfacility	MWh/year	30-November-2024 to 31-December-2024	39,813	The value is estimated quantity of net electricity generation supplied by the project Solar plant to the grid in year y. The electrical meters on site will be calibrated every five years in accordance with SANS 474 (national standard for calibration) and form part of the planned maintenance schedule for both checking and calibration. There is also a
		01-January-2025 to 31-December-2025	478,316	
		01-January-2026 to 31-December-2026	478,316	
		01-January-2027 to 31-December-2027	478,316	
		01-January-2028 to 31-December-2028	478,316	
		01-January-2029 to 31-December-2029	478,316	
		01-January-2030 to 31-December-2030	478,316	
		01-January-2030 to 29-November-2031	438,503	
		Total	3,348,215	

			backup meter both on site and at the Noko switching station which is used as an additional failsafe for QA/QC purposes.
$PE_{FC,y}$	tCO2/yr	0	Fuel usage for the emergency backup generator is monitored and converted to tCO2 based on the emission factor for the fuel used. However, Only monitored for record - keeping purposes, to confirm that the backup generator is only used for emergency backup.
PE_y	tCO2/yr	0	For all renewable energy power generation project activities, emissions due to the use of fossil fuels for the backup generator can be neglected.
ER_y	tCO2/yr	478,316	The value will be calculated

			ex-post using equation 09 of the applied methodology.
--	--	--	---

Detailed responsibilities and authorities for project management, monitoring procedures, calibration procedures and QA/QC procedures have been presented and were validated during follow up interviews. The detailed monitoring practice is considered appropriate and the implementation of these will enable subsequent verification of the project's emission reductions.

All relevant data will be archived electronically and further maintained for the entire crediting period plus two years. Based on the above assessment the validation team concludes that the PP is capable of implementing the monitoring plan and hence confirms compliance of VCS guidelines and the applied methodology .

3.4 Non-Permanence Risk Analysis

Not applicable.

4 VALIDATION OPINION

4.1 Validation Summary

Anthesis B.V. has commissioned Carbon Check (India) Private Ltd. (CC IPL) to validate the project activity 'Redstone CSP Project', with regard to VCS Version 4 requirements and the information provided by the project proponent related to the project design, planned operation, monitoring and reporting.

CC IPL has reviewed the project description documents and subsequently carried out on-site visit interviews to confirm the fulfilment of stated criteria. The project intends to reduce GHG emissions by displacing emission intensive grid electricity. A risk-based approach has been followed to perform this validation. In the course of validation, 04 CARs and 11 CLs are raised which have been resolved by the PP.

The project activity has applied the baseline and monitoring methodology, ACM0002, version 22.0: "Grid-connected electricity generation from renewable sources", which is an approved methodology under the CDM programme and is acceptable under VCS Version 4. The baseline has been determined in accordance with the stated approved baseline methodology.

Analysis of the proposed project activity reveals that the emission reductions resulting from the project activity are real, measurable and give long term benefits and are additional to what would have occurred in the absence of the project activity. The annual average emission reductions from the project activity are estimated to be 478,316 tCO₂e per annum. The emission reductions

forecast has been checked and is deemed likely that the stated amount is achieved given that the underlying assumptions do not change.

The monitoring plan makes sufficient provision for monitoring relevant project and baseline emission indicators. Responsibilities and authorities for project management, monitoring and reporting and QA/QC procedures have also been addressed.

Based on the information provided by the project developer, it is CCIPL's opinion that the project, "Redstone CSP Project" in South Africa as described in the VCS PD, Version 2.2.1 dated 12-August-2025, meets all relevant VCS Version 4 requirement and correctly applied approved CDM baseline and monitoring methodology ACM0002, version 22.0.

4.2 Validation Conclusion

Analysis of the proposed project activity reveals that the emission reductions resulting from the project activity are real, measurable and give long term benefits and are additional to what would have occurred in the absence of the project activity. The annual average emission reductions from the project activity are estimated to be 478,316 tCO₂e per annum. The emission reductions forecast has been checked and is deemed likely that the stated amount is achieved given that the underlying assumptions do not change.

Crediting period: From 30-November-2024 to 29-November-2031

Validated estimated GHG emission reductions and carbon dioxide removals for the project crediting period:

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU's (tCO ₂ e)	Estimated removal VCU's (tCO ₂ e)	Estimated total VCU's (tCO ₂ e)
30-Nov-2024 to 31-Dec-2024	39,813	0	0	39,813	0	39,813
01-Jan2025 to 31-Dec-2025	478,316	0	0	478,316	0	478,316
01-Jan2026 to 31-Dec-2026	478,316	0	0	478,316	0	478,316
01-Jan2027 to 31-Dec-2027	478,316	0	0	478,316	0	478,316

01-Jan2028 to 31-Dec-2028	478,316	0	0	478,316	0	478,316
01-Jan2029 to 31-Dec-2029	478,316	0	0	478,316	0	478,316
01-Jan2030 to 31-Dec-2030	478,316	0	0	478,316	0	478,316
01-Jan2031 to 29-Nov-2031	437, 947	0	0	437, 947	0	437, 947
Total	3, 348, 215	0	0	3, 348, 215	0	3, 348, 215

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

No commercial information is included.

APPENDIX 2.1 : REFERENCE DOCUMENTS

Item	GHG consideration
A. 1	Project Description version 1.5 dated 12-August-2024 Project Description version 2.0 dated 21-November-2024 Project Description version 2.0 dated 12-December-2024 Project Description version 2.1 dated 20-March-2025 Project Description version 2.2.1 dated 12-August-2025
A. 2	Evidence of project start date (A declaration may be provided in case of expected start date)
A. 3	Project implementation schedule
A. 4	Declarations from PP that the project activity is not claiming any other environmental credits other than VERs.
A. 5	Single line diagram and meter location layout
A. 6	Evidence of grievance mechanism
A. 7	Evidence of local stakeholder consultation
A. 8	South Africa's Integrated Resource plan (IRP 2019)
A. 9	ESKOM's annual integrated reports (2023) and records published on Eskom's website for each power station.
A. 10	Contract between VVB and PP
A. 11	VCS background documents • VCS Program Guide (v4.5) • VCS Standard (v4.7) • Program Definitions (v4.5) • Registration & Issuance Process (v4.5) • VCS Validation and Verification Manual (v 3.2)
A. 12	CDM background documents • CDM Methodology: ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0) • CDM Tool 01: Tool for the demonstration and assessment of additionality, Version 07.0.0 • CDM Tool 07: Tool to calculate the emission factor for an electricity system, Version 07.0 • CDM Tool 27: Investment analysis, Version 11.0

Item	Regulations and approval
B. 1	PPA signed between PP and Eskom
B. 2	Demonstration of legal ownership of project activity • IPP license • Lease direct agreement dated 06 December 2019 • Formalities certificate – Landowner
B. 3	Company registration certificates of each of the participating entities.
B. 4	EIA approval letter
B. 5	Demonstration of compliance with Occupational Health and Safety Act (Act No 85 of 1993) and Basic Conditions of Employment Act (No. 75 of 1997). (documented policy if any can be shared)

B. 6	Redstone's code of Conduct and Ethics Policy
B. 7	Evidence of Hazard rating 3 provided to the wastewater of project activity as per SABS 0228 code.
B. 8	Demonstration of compliance with day/night rating level of 70dBA as specified for industrial districts in SANS 10103. (if any)
B. 9	Approved EMP
B. 10	SANS 474 (to check calibration requirements of meters)
B. 11	Legal review and opinion on various contractual agreements. - Climate Legal - Anthesis_Verra_12082025_updated (1) - Legal review on another VCS project with similar public comment (Scatec - Opinion on Environmental Attributes (Execution)(1895646.1))
B. 12	Implementation agreement between PP and The Department Of Energy
B. 13	Industry Association letters : - Letter from SAWEA to IPP office - Letter from SACMA to IPP office
B. 14	Response letter from Redstone to IPP Office

Item	Technical design and technology
C. 1	Technical specifications of (including calibration requirements) : - Heliostats (demonstration of 93% reflectivity) - Receiver unit - Steam generation system - Storage unit - Power block - Emergency Digiset - Step up transformers - Electricity meter
C. 2	Document to justify operational and technical lifespan of the project.

Item	Financials
D. 1	Agreement between ACWA Power Solar Reserve Redstone Solar Thermal Power Plant (RF) (Pty) Ltd and Anthesis B.V. to demonstrate joint ownership of VERs generated through the project activity.

Item	Project operation
E. 1	QA/QC procedures
E. 2	Training manual, plan & records
E. 3	Operation and maintenance procedures

Item	Emission reduction
F. 1	ER calculation spreadsheet
F. 2	Grid Emission factor calculation sheet

APPENDIX 2.2 : BACKGROUND DOCUMENTS

Document
VCS Standard (v4.7)
VCS Program Guide (v4.5)
VCS Validation and Verification Manual version 3.2
Registration & Issuance Process (v4.5)
VCS Programme Definitions version 4.5
VCS PD template version 4.4
Applied baseline and monitoring methodology, ACM0002 Version 22.0
CDM Tool 01: Tool for the demonstration and assessment of additionality, Version 07.0.0
CDM Tool 07: Tool to calculate the emission factor for an electricity system, Version 07.0
CDM Tool 27: Investment analysis, Version 11.0

APPENDIX 2.3 : CERTIFICATES OF COMPETENCE

Carbon Check (India) Private Limited

Certificate of Competency

Campal Deepak Kadam

has been qualified as per CCIPL's internal qualification procedures in accordance with the requirements of CDM AS, A 6.4 AS/ ISO/IEC14065:2020, ISO/IEC 17029:2019 and other applicable GHG programs:

for the following functions and requirements:

- | | | | |
|--|---|--|---|
| ☐ Management Function | ☐ Administrative | ☒ Validator/Verifier | ☒ Team Leader |
| ☐ Acting Team Leader | ☒ Technical Expert | ☐ Technical Reviewer | ☐ Trainee Assessor |
| ☒ Assessor | ☒ Regional Expert | ☐ Gender Expert | ☐ Financial Expert |
| <div style="border: 1px solid black; padding: 2px; display: inline-block;">India</div> | | | |
| ☐ Plastic Waste Expert | ☐ Health Expert | ☐ CCB Expert | ☒ SDG Expert |
| ☒ Expert Social Aspect | ☒ Expert Environmental Aspect | ☐ Internal Auditor | ☐ Legal Expert |
| ☒ FOEN Approved Technical Expert | ☐ FOEN Approved Quality Officer | | |

in the following Technical Areas:

- ☐ PWRP 2 - Plastic Waste Collection & Recycling
- ☒ TA 1.1 - Thermal energy generation
- ☒ TA 1.2 - Renewable Energy generation
- ☐ TA 10.1 - Fugitive emissions from oil and gas
- ☒ TA 13.1 - Solid waste and Wastewater
- ☒ TA 13.2 - Manure
- ☐ TA 14.1 - Afforestation and reforestation (except REDD & REDD+)

- ☐ TA 14.1 - Afforestation and reforestation (only REDD & REDD+)
- ☐ TA 15.1 - Agriculture (With Model Validation)
- ☐ TA 15.1 - Agriculture (Without Model Validation)
- ☐ TA 16.1 Carbon Capture and Storage ☒ TA 2.1 - Energy Distribution
- ☐ TA 3.1 - Demand Side Energy Efficiency
- ☐ TA 4.1 - Cement and Lime Production
- ☐ TA 4.n - Waste Heat Recovery & Fuel Switch ☐ TA 5.1 - Chemical Industry
- ☐ TA 5.2 - Caprolactam, Nitric and Adipic Acid ☐ TA 7.1 - Transport
- ☐ TA 8.1 - Mining/Mineral Production
- ☐ TA 9.1 - Aluminium and magnesium production
- ☐ TA 9.2 - Iron, Steel and Ferro-alloy Production
- ☐ TA PWRP 1 - Plastic Waste Collection & Recycling

Issue Date 26-11-2025 **Expiry Date** 26-11-2026

Vikash Kumar Singh
Director - Compliance

Carbon Check (India) Private Limited

Certificate of Competency

Anubhav Dimri

has been qualified as per CCIPL's internal qualification procedures in accordance with the requirements of CDM AS, A 6.4 AS/ ISO/IEC14065:2020, ISO/IEC 17029:2019 and other applicable GHG programs:

for the following functions and requirements:

- | | | | |
|--|---|--|--|
| ☒ Management Function | ☒ Administrative | ☒ Validator/Verifier | ☒ Team Leader |
| ☐ Acting Team Leader | ☒ Technical Expert | ☒ Technical Reviewer | ☐ Trainee Assessor |
| ☒ Assessor | ☒ Regional Expert | ☐ Gender Expert | ☒ Financial Expert |
| | <div>India</div> | | |
| ☒ Plastic Waste Expert | ☐ Health Expert | ☒ CCB Expert | ☒ SDG Expert |
| ☒ Expert Social Aspect | ☒ Expert Environmental Aspect | ☒ Internal Auditor | ☐ Legal Expert |
| ☒ FOEN Approved Technical Expert | ☐ FOEN Approved Quality Officer | | |

in the following Technical Areas:

- ☐ PWRP 2 - Plastic Waste Collection & Recycling
- ☒ TA 1.1 - Thermal energy generation
- ☒ TA 1.2 - Renewable Energy generation
- ☐ TA 10.1 - Fugitive emissions from oil and gas
- ☒ TA 13.1 - Solid waste and Wastewater
- ☒ TA 13.2 - Manure
- ☒ TA 14.1 - Afforestation and reforestation (except REDD & REDD+)

- ☒ TA 14.1 - Afforestation and reforestation (only REDD & REDD+)
- ☒ TA 15.1 - Agriculture (With Model Validation)
- ☒ TA 15.1 - Agriculture (Without Model Validation)
- ☒ TA 16.1 Carbon Capture and Storage ☐ TA 2.1 - Energy Distribution
- ☒ TA 3.1 - Demand Side Energy Efficiency
- ☐ TA 4.1 - Cement and Lime Production
- ☐ TA 4.n - Waste Heat Recovery & Fuel Switch ☐ TA 5.1 - Chemical Industry
- ☐ TA 5.2 - Caprolactam, Nitric and Adipic Acid ☐ TA 7.1 - Transport
- ☒ TA 8.1 - Mining/Mineral Production
- ☐ TA 9.1 - Aluminium and magnesium production
- ☐ TA 9.2 - Iron, Steel and Ferro-alloy Production
- ☐ TA PWRP 1 - Plastic Waste Collection & Recycling

Issue Date 13-01-2026 **Expiry Date** 13-01-2027

Vikash Kumar Singh
Director - Compliance

Carbon Check (India) Private Limited

Certificate of Competency

Indumathi C

has been qualified as per CCIPL's internal qualification procedures in accordance with the requirements of CDM AS, A 6.4 AS/ ISO/IEC14065:2020, ISO/IEC 17029:2019 and other applicable GHG programs:

for the following functions and requirements:

- | | | | |
|--|---|--|--|
| ☐ Management Function | ☐ Administrative | ☒ Validator/Verifier | ☒ Team Leader |
| ☐ Acting Team Leader | ☒ Technical Expert | ☒ Technical Reviewer | ☐ Trainee Assessor |
| ☒ Assessor | ☒ Regional Expert | ☐ Gender Expert | ☒ Financial Expert |
| | India, Sri Lanka | | |
| ☒ Plastic Waste Expert | ☐ Health Expert | ☐ CCB Expert | ☒ SDG Expert |
| ☒ Expert Social Aspect | ☒ Expert Environmental Aspect | ☐ Internal Auditor | ☐ Legal Expert |
| ☒ FOEN Approved Technical Expert | ☐ FOEN Approved Quality Officer | | |

in the following Technical Areas:

- ☒ TA 1.1 - Thermal energy generation
- ☒ TA 1.2 - Renewable Energy generation
- ☐ TA 10.1 - Fugitive emissions from oil and gas
- ☒ TA 13.1 - Solid waste and Wastewater
- ☒ TA 13.2 - Manure
- ☐ TA 14.1 - Afforestation and reforestation (except REDD & REDD+)
- ☐ TA 14.1 - Afforestation and reforestation (only REDD & REDD+)

- ☐ TA 15.1 - Agriculture (With Model Validation)
- ☐ TA 15.1 - Agriculture (Without Model Validation)
- ☐ TA 16.1 Carbon Capture and Storage ☐ TA 2.1 - Energy Distribution
- ☒ TA 3.1 - Demand Side Energy Efficiency
- ☐ TA 4.1 - Cement and Lime Production
- ☐ TA 4.n - Waste Heat Recovery & Fuel Switch ☐ TA 5.1 - Chemical Industry
- ☐ TA 5.2 - Caprolactam, Nitric and Adipic Acid ☐ TA 7.1 - Transport
- ☐ TA 8.1 - Mining/Mineral Production
- ☐ TA 9.1 - Aluminium and magnesium production
- ☐ TA 9.2 - Iron, Steel and Ferro-alloy Production
- ☐ TA PWRP 1 - Plastic Waste Collection & Recycling

Issue Date 26-11-2025 Expiry Date 26-11-2026

Vikash Kumar Singh
Director - Compliance

APPENDIX 2.4 : FINDINGS LOG

Table 1. CLs from this validation

CL ID	01	Section no.	1.12	Date: 11/11/2024
Description of CL				
1. In the referenced section of the PD, PP has not specified the age and average lifetime of the equipment to be used in the project activity. <u>Background :</u> <i>Section 1.12 of the PD template v 4.4 states that, “. Include in the description information about the age and average lifetime of the equipment based on manufacturer’s specifications and industry standards.....”</i> 2. Further in the same section, PP has not clearly justified the types and levels of service as compared to the baseline scenario. <u>Background :</u> <i>Clearly explain how the same types and levels of services provided by the project would have been provided in the baseline scenario.</i> 3. Additionally, section 1.12 of the PD template v 4.4 states, “Where appropriate, provide a list of facilities, systems, and equipment in operation under the existing scenario prior to the implementation of the project.” However, PP has not specified/differentiated and sufficiently described the equipments that were installed at the time of validation. (e.g. nameplate, serial number and other technical details of power block and steam turbine generator etc.). 4. PP has provided implementation status of the project. However, PP has not clearly specified key milestones such as expected listing on VERRA registry, expected completion of validation and first verification.				
PP response				Date: 11/11/2024

Section 1.12 of the PD was updated to specify the equipment age and lifetime as per the PPA and technology specifications.

Section 1.12 of the PD was updated to specify the types and levels of services supplied by the project, and to demonstrate that it is similar to the baseline scenario.

Section 1.12 of the PD was updated to list the facilities in operation under the existing scenario prior to the implementation of the project.

Section 1.12 was updated to include additional details on the SGS and power block systems

Section 1.12 was updated to include key milestones of the validation / verification process.

Documentation provided by PP

[VCS PD 5146 18NOV2024 v2 Tracked Changes.docx](#)

[VCS PD 5146 18NOV2024 v2.pdf](#)

VVB assessment

Date: 02/12/2024

1. PP has updated section 1.12 of the PD to clearly specify the equipment age and lifetime. VVB has cross checked the reported information with PPA and technology specifications. Hence, this part of the finding is closed.

2. PP has updated section 1.12 of the PD to clearly demonstrate that types and service levels in project scenario are similar to baseline scenario. Hence, this part of the finding is closed.

3. PP has updated section 1.12 of the PD to clearly specify the equipments in operation at the time of validation. Hence, this part of the findings is closed.

4 PP has updated section 1.12 of the PD to include relevant milestones. Hence, this part of the finding is closed.

CL ID	02	Section no.	2.1.2	Date:	11/11/2024
Description of CL					
In the referenced section of PD, PP has not sufficiently described feedback received during stakeholder engagement. Further there is not enough information regarding their addressal and justification for no action required as specified in PD.					
<u>Background</u> Section 2.1.1 of the PD template v 4.4, row 'stakeholder input' states that, "Describe how due account was taken of all input received during the consultation. Include details on any updates to the project design or justify why updates were not necessary or appropriate. "					
PP response					Date:
Section 2.1.2 was updated to include comments received from stakeholders, and to justify why no updates to the project design were necessary					12/11/2024
Documentation provided by PP					
VCS PD 5146 18NOV2024 v2 Tracked Changes.docx VCS PD 5146 18NOV2024 v2.pdf					
VVB assessment					Date:
					02/12/2024

PP has updated section 2.1.2 of the PD to describe the stakeholder interaction and addressal of stakeholder inputs. Hence, this finding is closed.

CL ID	03	Section no.	1.8	Date: 11/11/2024
Description of CL				
In the referenced section of the PD, PP states that, <i>“Anthesis and ACWA Power Solar Reserve Redstone Solar Thermal Power Plant (RF) (Pty) Ltd have a legal agreement in place for the joint ownership of a registered carbon project.”</i>				
PP to provide the document for VVB’s review.				
PP response				Date: 11/11/2024
The document "CL3_Redstone_Termsheet_signed.pdf" was uploaded into the shared folder as evidence. This is a legally binding agreement between the project proponents and addresses the terms for the carbon credit development.				
Documentation provided by PP				
CL3_Redstone_Term_Sheet_signed.pdf				
VVB assessment				Date: 02/12/2024
PP has provided the requested document which is reviewed by VVB and is deemed acceptable. Hence, this finding is closed.				

CL ID	04	Section no.	Start date evidence	Date: 11/11/2024
Description of CL				
PP has provided evidence titled ‘A.2_Evidence of Start Date’ for start date of project activity. PP to clarify how the evidence is in line with definition of start date in VCS. (In case of future start date a signed declaration stating that the project is expected to begin on the said date).				
PP response				Date: 12/11/2024
Redstone CSP is currently in the commissioning phase of the project and the expected commercial operation date (C.O.D.) for the project is 30 November 2024. This will be confirmed with a formal sign-off in accordance with the PPA in the future. This sign off will then be the start date of the project. Letter "CL4_Redstone COD Expectation.pdf" uploaded.				
Documentation provided by PP				
CL4_Redstone COD Expectation.pdf				
VVB assessment				Date: 02/12/2024
VVB has assessed the declaration provided by PP and confirms that the expected start of project activity is 30 th November 2024. Hence, this finding is closed.				

CL ID	05	Section no.	--	Date: 11/11/2024
Description of CL				
In the document issued by EA, Dept. of environmental affairs, the project is titled 'Humansrus CSP'. PP to clarify regarding the same and provide an amended copy if any.				
PP response				Date: 12/11/2024
The Redstone CSP project was initially developed by SolarReserve, a California-based company, and it was originally named "Humansrus CSP". The project was expected to achieve financial close by the end of 2015 and begin operations in early 2018. However, it faced multiple delays due to issues with South Africa's renewable energy permitting process and financial challenges. After SolarReserve faced financial difficulties, ACWA Power took over the project. In 2021, ACWA Power restarted the project with the help of POWERCHINA subsidiary SEPCOIII, which has extensive experience in constructing CSP projects. The project was renamed "Redstone CSP". The initial Environmental Authorization (EA), DEA Reference No: 12/12/20/2316, issued on 06 August 2012 has been amended a number of times and the first amendment, granted on 26 June 2013, authorises the name change from 'Humansrus Solar Thermal Power Plant (Humansrus CSP)' to 'Redstone Solar Thermal Power Plant'. While the fourth amendment, DEA Reference No: 12/12/20/2316/AM4, granted on 27 March 2015, authorises the name change from 'Redstone Solar Thermal Power Plant' to 'ACWA Power SolarReserve Redstone Solar Thermal Power Plant'. Copies of the EA and amendments are attached.				
Documentation provided by PP				
CL5 EA Ammendment Name Change.pdf CL5 EA Ammendment 2nd name change.pdf CL5 Statement Name change.pdf				
VVB assessment				Date: 02/12/2024
VVB has assessed the copies of EA and amendments and deems that the clarification provided by PP is acceptable. Hence, this finding is closed.				

CL ID	06	Section no.	4.1	Date: 11/11/2024
Description of CL				
In section 4.1.3, PP has not clearly demonstrated the share of LCMR in the last five years (average) to be less than 50%. Moreover, in the grid emission factor calculation sheet, share of LCMR for each year is calculated separately and equation 3, tool 07 is not applied to determine average.				
PP response				Date: 11/11/2024
Section 4.1.3 was updated to demonstrate that the average share of LCMR plants over the last 5 years has been 19.9 %. This has also been updated in the GEF calculation sheet.				
Documentation provided by PP				

[VCS PD 5146 18NOV2024 v2 Tracked Changes.docx](#)

[VCS PD 5146 18NOV2024 v2.pdf](#)

[F.1_F.2 Redstone Grid Emission Factor_ER.xlsx](#)

VVB assessment

Date: 02/12/2024

PP has updated section 4.1 of the PD to clearly demonstrate that the share of LCMR in last five years is less than 50%. VVB has cross-checked the calculation with equation 3 of TOOL 07 and deemed appropriate. Hence, this finding is closed.

CL ID	07	Section no.	5.2	Date:	11/11/2024
Description of CL					
In the data/parameter table ' <i>EGfacility,y</i> ' , PP has not provided the details of monitoring equipment i.e. electricity meter make, serial number, accuracy, class.					
Additionally, details of backup meter are not provided.					
PP response				Date:	12/11/2024
Section 5.2 of the PD has been updated to include the meter make, serial number, class and accuracy. Supporting documents include photographs of all meters.					
Documentation provided by PP					
CL7 Photos of Meters.pdf					
VVB assessment				Date:	02/12/2024
PP has updated section 5.2 of the PD to include details of meter make, serial number, class and accuracy. VVB has cross-checked the same and deemed appropriate. Hence, this finding is closed.					

CL ID	08	Section no.	ER sheet	Date: 11/11/2024
Description of CL				
The estimated emission reductions reported in the PD do not match the estimated emission reductions calculated in the ER sheet. Further vintage wise ER estimation is not provided in the ER sheet. Further clarification is required.				
PP response				Date: 14/11/2024
The PD has been updated to reflect emission reductions that correspond with the ER spreadsheet. Furthermore, the ER spreadsheet has been updated to reflect the vintage wise ER estimations				
Documentation provided by PP				
VCS PD 5146 18NOV2024 v2 Tracked Changes.docx				
VCS PD 5146 18NOV2024 v2.pdf				
F.1_F.2 Redstone Grid Emission Factor_ER.xlsx				
VVB assessment				Date: 02/12/2024

PP has updated the PD to report consistent values of estimated emission reductions. Additionally, vintage wise breakdown is provided. Hence, this finding is closed.

CL ID	09	Section no.	1.1	Date: 11/11/2024
Description of CL				
In the referenced section of PD, PP states that, <i>"This enables the plant to store thermal energy for a period of up to 12 hours, and hence it can provide electricity to the equivalent of more than 200,000 South African homes during peak demand periods via the national grid, including after the sun has set."</i> However, during interview with the plant manager, VVB has noted that the CSP plant will not export electricity to the grid after 10.00 PM. In light of this note, PP is required to clarify above statement in PD.				
PP response				Date: 14/11/2024
Section 1.1 was updated to specify more clearly that: The plant can store thermal energy for up to 12 hours as per the design, but it will supply electricity to the national grid each day from 05:00 to 22:00, as per the PPA, which amounts to 17 hours.				
Documentation provided by PP				
VCS PD 5146 18NOV2024 v2 Tracked Changes.docx VCS PD 5146 18NOV2024 v2.pdf CL9_Power Purchase Agreement - Redstone.pdf CL9_First Amendment to Power Purchase Agreement (Executed).pdf				
VVB assessment				Date: 02/12/2024
PP has updated section 1.1 of the PD to clarify regarding plant operations. PP has specified that plant can store thermal energy for up to 12 hours as per the design, but it will supply electricity to the national grid each day from 05:00 to 22:00, as per the PPA, which amounts to 17 hours. Hence, this finding is closed.				

CL ID	10	Section no.	1.12	Date: 11/11/2024
Description of CL				
In the referenced section of PD, PP has specified that there will be two emergency diesel generators on the plant site. However, during interviews with the plant management, VVB has noted that there will be one emergency diesel generator. PP to clarify the number of diesel generator and make necessary corrections.				
PP response				Date: 21/11/2024
Section 1.12 of the PD was updated to state that there is one emergency diesel generator on site. Supporting documents provided				
Documentation provided by PP				

CL10_EMERGENCY DIESEL GENERATORS DATASHEET.pdf	
CL10_Emergency Diesel Generators System Calculation Report.pdf	
VVB assessment	Date: 02/12/2024
PP has clarified that there will be one generator on plant site and updated the sae information in the PD. Hence, this finding is closed.	

CL ID	11	Section no.	1.1	Date: 11/11/2024
Description of CL				
In the referenced section of PD, PP states that, “Thermal energy is converted to electricity through 115MW high temperature subcritical steam turbine generating units.”				
However, PP also states in the PD that the CSP plant is nominal 100 MW plant. PP is required to clarify how the capacity of the plant is determined.				
PP response				Date: 14/11/2024
Section 1.1 of the PD was updated to state more clearly that the maximum capacity of the turbine generator is 115 MW, which corresponds to 100% load. However, the operating as per PPA condition will be 100 MW, which corresponds to 87% load.				
Documentation provided by PP				
VCS PD 5146 18NOV2024 v2 Tracked Changes.docx				
VCS PD 5146 18NOV2024 v2.pdf				
CL11_RDST-01-MKA-EHP-SIE-00057_Generator data sheet.pdf				
VVB assessment				Date: 02/12/2024
PP has clarified that as per PPA, the operation condition is 100 MW corresponding to 87%. VVB has cross-checked the same with PPA and deemed appropriate. Hence, this finding is closed.				

Table 2. CARs from this validation

CAR ID	01	Section no.	3.3	Date: 11/11/2024
Description of CAR				
In the referenced section of PD, the GHG sources identified by PP are not in line with the applied methodology. PP is required to make necessary corrections.				
PP response				Date: 11/11/2024
The table in section 3.3 of the PD was updated to account for CO ₂ emissions from fossil fuel consumption in the project scenario.				
Documentation provided by PP				
VCS PD 5146 18NOV2024 v2 Tracked Changes.docx				
VCS PD 5146 18NOV2024 v2.pdf				
VVB assessment				Date: 02/12/2024

PP has revised section 3.3 of the PD to specify the GHG sources in line with applied methodology. Hence, this finding is closed.

CAR ID	02	Section no.	4.4	Date: 11/11/2024
Description of CAR				
Editorial corrections : 1. PP is required to use international standard while indicating decimal numbers. 2. 'Error! Reference source not found.' is seen across the PD. PP to rectify the same. 3. In section 5.1 'EF_{grid,OMsimple},' is repeated. 4. PP to name the PD file as per the template requirement. (For projects requesting registration approval: VCS PD ProjectID DDMMMYYYY) 5. PP to provide link of grievance portal on Anthesis website on pg 37.				
PP response				Date: 11/11/2024
1. Commas were inserted into all figures to separate thousands and to correspond with international decimal standards. 2. All reference errors in the document were removed. 3. The duplicate reference to 'EF_{grid,OMsimple}' was removed from section 5.1 4. The document was renamed to " VCS PD 5146 18NOV2024 v2" 5. The link to the Anthesis grievance portal was inserted in section 2.1.4 of the PD				
Documentation provided by PP				
VCS PD 5146 18NOV2024 v2 Tracked Changes.docx VCS PD 5146 18NOV2024 v2.pdf				
VVB assessment				Date: 02/12/2024
PP has made the necessary editorial corrections. Hence, this finding is closed.				

CAR ID	03	Section no.	1.3	Date: 11/11/2024
Description of CAR				
In the referenced section of PD, PP specifies the project type as 'Project'. However, this is not a type in VCS. PP to make necessary corrections.				
PP response				Date: 11/11/2024
Updated section 1.3 to correctly identify Renewable energy type project				
Documentation provided by PP				
VCS PD 5146 18NOV2024 v2 Tracked Changes.docx VCS PD 5146 18NOV2024 v2.pdf				
VVB assessment				Date: 02/12/2024
PP has revised section 1.3 of the MR to specify the project type which is deemed appropriate. Hence, this finding is closed.				

CAR ID	04	Section no.	2.1.5	Date: 11/11/2024
Description of CAR				
In the referenced section of PD, PP to complete the section as per template guideline as public commenting period has lapsed.				
PP response				Date: 12/11/2024
Section 2.1.5 of the PD was updated to state that no comments were received during the global stakeholder public commenting period. Additional comments and inputs from local stakeholders were also included in section 2.1.2 of PD.				
Documentation provided by PP				
VCS PD 5146 18NOV2024 v2 Tracked Changes.docx VCS PD 5146 18NOV2024 v2.pdf				
VVB assessment				Date: 02/12/2024
PP has updated section 2.1.5 of the PD to specify that no comments were received during the public consultation period. Hence, this finding is closed.				

Table 3. FARs from this validation

FAR ID	01	Section no.	3.2.1.5	Date: 19/01/2026
Description of CAR				
The VVB carrying out the verification shall assess and confirm whether any changes occurred to followings: - the Implementation A2greement (IA); - the Power Purchase Agreement (PPA); and/or - applicable national laws, regulations, or government directives that may affect the ownership, registration, or monetisation of environmental attributes, including carbon credits. Further, VVB shall explicitly document the outcome of the assessment in the verification report, including: - confirmation that no such changes have occurred; or - where changes have occurred, a clear assessment of their implications for the project's continued eligibility under the VCS Program.				